5.2. Four - sided Figures

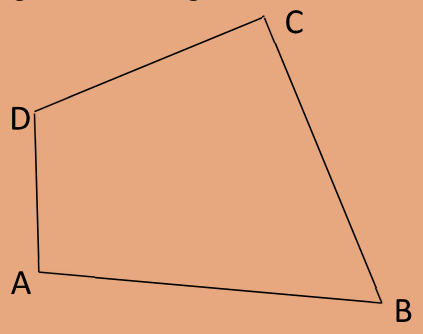
At the end of this section you should be able to:

- *construct and describe properties of four-sided figures*

Activity 4.2.1

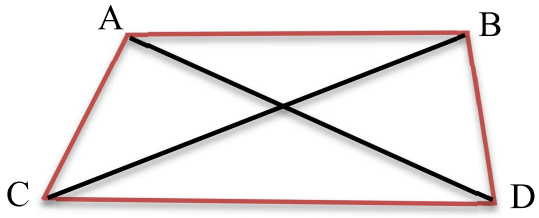
Discuss with your friends/partners:

1. considering the geometric figure below and answer the following question



a. The geometric figure has _____ sides
b. Name all its sides.
c. Name all its vertices.
d. Name all its interior angles.
e. The geometric figure is called_____.
f. Name the geometric figure using the four vertices.
g. The vertex opposite to D is _____
h. Name the adjacent vertex to D.
i. Name the opposite side to $\overline{AD}$.
j. Name the adjacent side to $\overline{AD}$.

Definition 5.2: A quadrilateral is a four-sided geometric figure bounded by line segments.



- A quadrilateral is named by using its four vertices in clock wise or anti clock wise direction.
- The above quadrilateral is named as quadrilateral ACDB or BACD or CDBA and so on, but it cannot be named as ABCD or ADCB.
- The sides of the quadrilateral ABDC above are $\overline{AB}, \overline{AC}, \overline{CD}, and\ \overline{BD}$

i. Opposite side to $\overline{AB}$ is $\overline{CD}$

ii. Opposite side to $\overline{AC}$ is $\overline{BD}$

iii. The adjacent sides to $\overline{AB}$ are $\overline{AC}$ and $\overline{BD}$

- The point where the sides of a quadrilateral meets is called Vertex (plural form vertices). The vertices of the quadrilateral ABDC above are A, B, D and C.
  i. The opposite vertex to A is D.
  ii. The vertex adjacent to A are C and B.
- The angles formed by adjacent side of the quadrilateral are interior angles of the quadrilateral. The interior angles of quadrilateral ABDC above are $\angle A, \angle B, \angle D$ and $\angle C$
  i. The opposite angle to $\angle A$ is $\angle D$.
  ii. The angle adjacent to $\angle A$ are $\angle C$ and $\angle B$.
- A line segment that connects two opposite vertices of the quadrilateral is called diagonal. The diagonals of quadrilateral ABDC above are $\overline{AD}$ and $\overline{CB}$.

Construction and properties of trapezium

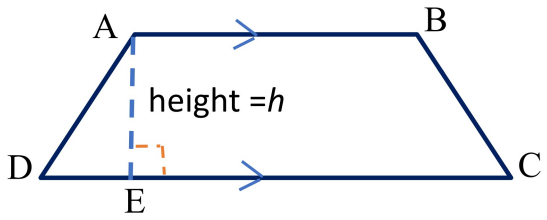
Activity 5.2.2

Discuss with your friends/partners:

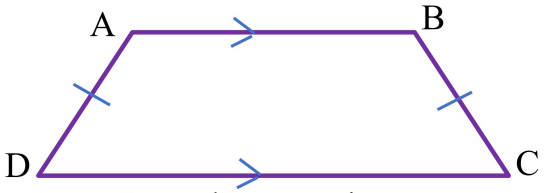
Consider the quadrilateral below and answer the following questions

1. Which sides of this quadrilateral are parallel?
2. What do we call such type of quadrilateral?
3. Measure $\angle A, \angle B, \angle D$ and $\angle C$ and calculate
   i. $\angle A + \angle D =$ ____
   ii. $\angle C + \angle B =$ ____
   iii. $\angle A + \angle B + \angle C + \angle D =$ ____

Definition 5.3: A trapezium is a special type of quadrilateral in which exactly one pair of opposite sides are parallel.



- The parallel sides of trapezium are called the bases of trapezium. In the above trapezium, the parallel sides $\overline{AB}$ and $\overline{DC}$ are bases.
- The distance between the bases is called the height (or altitude) of the trapezium. $\overline{AE}$ is the height.
- The non-parallel sides of the trapezium are called legs of the trapezium. The non-parallel sides $\overline{AD}$ and $\overline{BC}$ are legs.
- If the legs of trapezium are congruent, then trapezium is called isosceles trapezium.

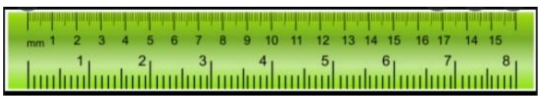


Isosceles trapezium

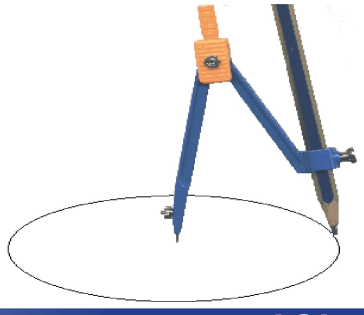
Construction of trapezium

- To perform geometric construction, you need tools like:
  - Ruler
  - Compass
  - Protractor

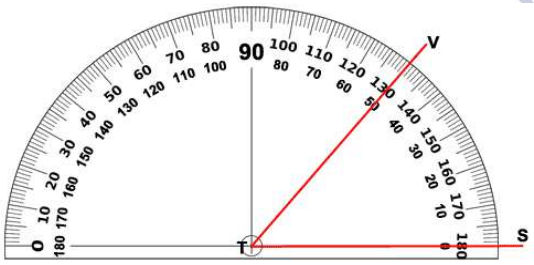
Ruler is used to construct line or line segments.



Compass is used to construct circles or arcs.



Protractor is used to measure or construct angles.

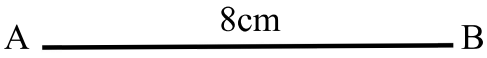


Example 1: construct a trapezium ABCD using the information given below.

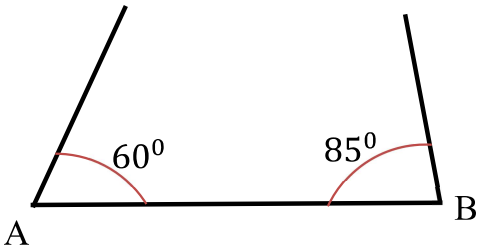
Given: $\overline{AB} \parallel \overline{CD}$, $AB = 8cm, BC = 5cm, m(\angle A) = 60^0, m(\angle B) = 85^0$

Solution:

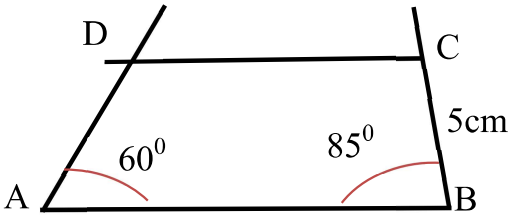
Step 1: draw a line segment $AB = 8cm$



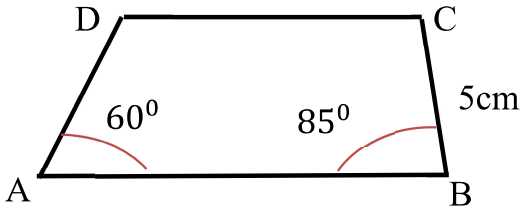
Step 2: construct $m(\angle A)$ *and* $m(\angle B)$ with the given measures.



Step 3: Mark point C on the side of $\angle B$ such that $BC = 5cm$

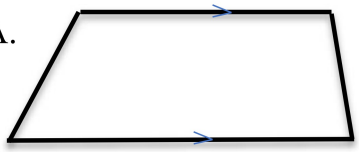


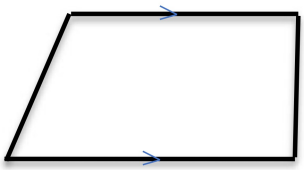
Step 4: Draw a line through C parallel to AB, intersecting the side of $\angle A$ at D

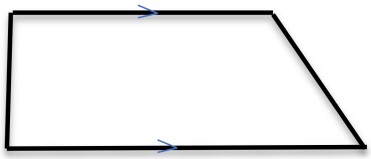


Exercise 5.2.1

1. Fill in the blank space with the correct answer.
   a. Four-sided geometric figure is called_____
   b. A line segment that joins opposite vertex of a quadrilateral is _____
   c. The point where the sides of a quadrilateral meet is called_____
   d. The tools used for geometric construction are _____, _____ and____
2. Which one of the following is **not** a trapezium?

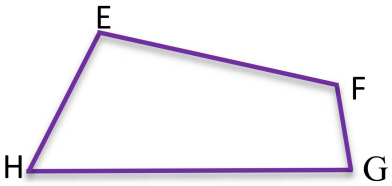
A. 

C. 

B. 

D.

3. Considering the quadrilateral below answer the following questions

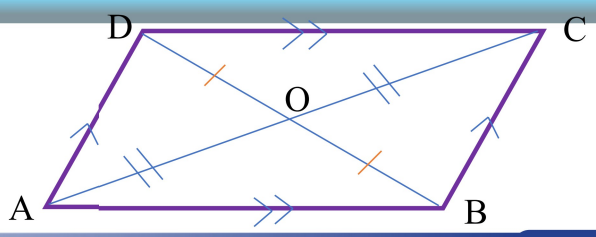


   a. the opposite side to FG is _____
   b. the adjacent sides to HG are___, ____
   c. the opposite angle to $\angle F$ is ____
   d. The quadrilateral is named as _____
   e. the diagonal of the quadrilateral are line segments joining ___ and___ or ___and_____
4. Construct a trapezium EFGH using the information given below.

   Given: $\overline{EF} \parallel \overline{GH}$, $EF = 10cm, FG = 6\ cm, m(\angle E) = 30^0, m(\angle F) = 80^0$

Construction and properties of parallelogram

Definition 5.4: A parallelogram is a quadrilateral in which both pair of opposite sides are parallel.



In the figure $\overline{AB} \parallel \overline{CD}$ *and* $\overline{AD} \parallel \overline{BC}$, thus ABCD is a parallelogram.

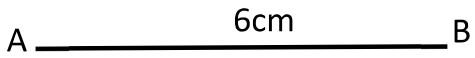
Construction of parallelogram

- Construct a parallelogram ABCD using the information given below

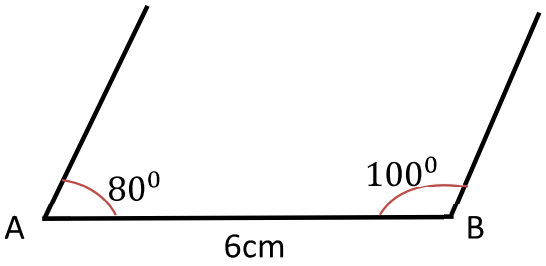
Given: $\overline{AB} \parallel \overline{CD}$ *and* $AB = 6cm, BC = 4cm, m(\angle A) = 80^0, m(\angle B) = 100^0$

Solution:

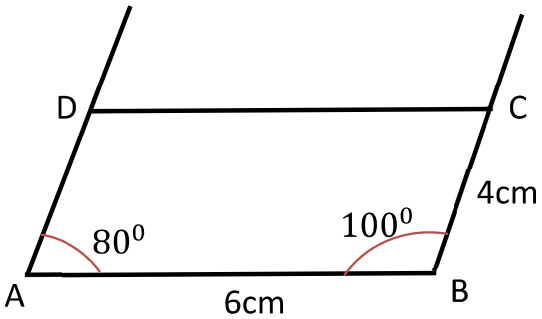
Step 1: Draw a line segment $AB = 6cm$



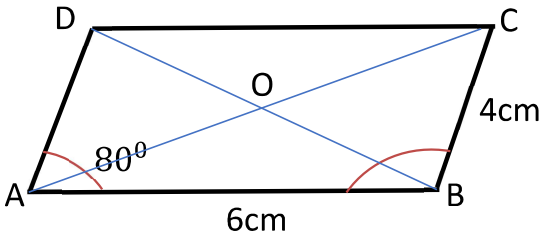
Step 2: construct $\angle A$ *and* $\angle B$ so that $(\angle A) = 80^0$ *and* $m(\angle B) = 100^0$



Step 3: mark point C on side of $\angle B$ such that $BC = 4cm$. And draw a line through C parallel to AB so that it meets the side of $\angle A$ at point D.



Finally, the parallelogram ABCD satisfying the given condition is



Using the constructed parallelogram above answer the following questions.

1. Measure $\angle D$ and $\angle C$. compare $\angle D$ *with* $\angle B$, and $\angle A$ *with* $\angle C$

From the result in 1:

- Opposite angles of a parallelogram are _____

2. Find the value of

i. $m(\angle A) + m(\angle B) =$ _____ iii. $m(\angle C) + m(\angle D) =$ _____

ii. $m(\angle B) + m(\angle C) =$ _____ iv. $m(\angle A) + m(\angle D) =$ _____

From the result in 2:

- Consecutive (or adjacent) angles of a parallelogram are _____

3. Measure side AD and DC. Compare AD with BC and AB with DC.

From the result in 3:

- Opposite sides of a parallelogram are _____

4. Construct diagonals AC and DB, so that they intersect at O. Compare the length of DO with OB and AO with OC.

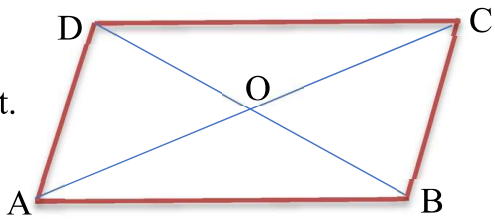
From the result in 4:

- The diagonals of parallelogram _____ each other.

From the activity above the properties of parallelogram is summarized as follows:

1. Opposite sides of parallelogram are congruent.

$$i.e.\ AD = BC\ and\ AB = DC$$



2. Opposite angles of a parallelogram are congruent.

$$i.e.\ m(\angle A) = m\ (\angle C)\ and\ m(\angle B)$$
$$= m\ (\angle D)$$

3. Consecutive (or adjacent) angles of a parallelogram are supplementary.

$$i.e.\ m(\angle A) + m(\angle B) = 180^0, m(\angle B) + m(\angle C) = 180^0$$

4. The diagonals of parallelogram bisect each other.

$$i.e.\ AO = OC, and\ DO = OB$$

Note:

Bisect means "divides exactly in to two equal parts.

Supplementary angles are angles whose sum is 180^0.

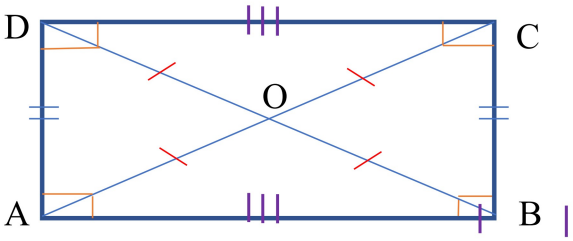
Congruent means equal in length.

Construction and properties of special parallelograms

Rectangle, Rhombus and square are special parallelograms. They all satisfy the properties of parallelogram.

A. Rectangle

Definition 5.5: A rectangle is a parallelogram with all its angles are right angle.



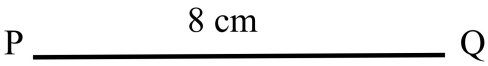
Construction of Rectangle

- Construct a rectangle PQRS using the information given below

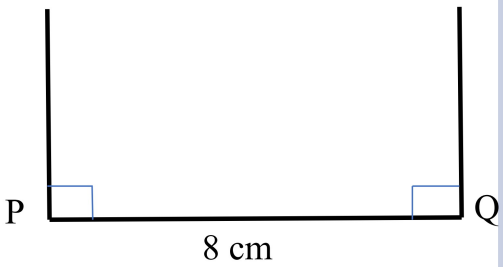
Given: $\overline{PQ} \parallel \overline{RS}, PQ = 8cm, QR = 4cm, m(\angle P) = 90^0$

Solution:

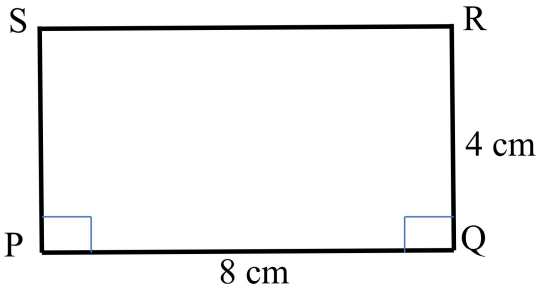
Step 1: construct a line segment PQ with length 6 cm.



Step 2: construct $m(\angle P) = 90^0$ *and* $m(\angle Q) = 90^0$



Step 3: mark point R and S, such that $PS = QR = 4cm$ and connect RS parallel to PQ



Using the constructed rectangle above answer the following questions.

1. Does it satisfy all properties of parallelogram?
2. Compare the diagonals PR and SQ.

The properties of rectangle are summarized as follows:

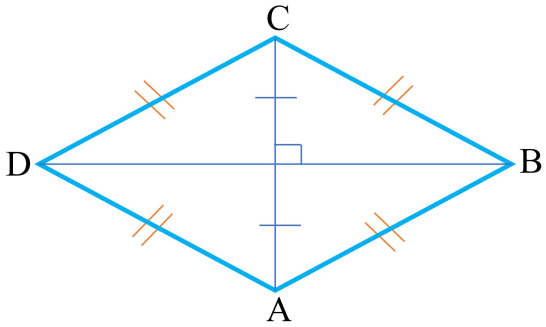
1. Rectangle satisfies all properties of parallelogram.
2. The diagonals of rectangle are equal in length and bisect each other.
3. All angles of rectangle are right angle.

Note:

1. Right angle is an angle that measures 90^0
2. All rectangles are parallelogram, but all parallelograms are not rectangles
3. A quadrilateral with congruent diagonals is not necessarily rectangle
4. A parallelogram with congruent diagonals is rectangle.

B. Rhombus

Definition 5.6: A rhombus is a parallelogram in which all its sides are congruent.



Properties of rhombus

i. All properties of parallelogram are properties of rhombus.

ii. All sides of rhombus are congruent ($i.e. AB = BC = CD = AD$)

iii. The diagonals of rhombus are perpendicular to each other (i.e. $AC \perp DB$)

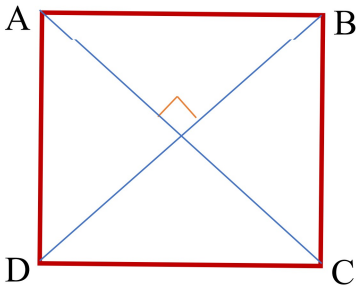
iv. The diagonals of rhombus bisects the angles at the vertices. For example, in the above rhombus $\angle CDA$ is bisected by the diagonal DB, so that $m(\angle CDB) = m(\angle ADB)$.

Note:

1. All rhombus are parallelogram, but all parallelograms are not rhombus.
2. The diagonals of rhombus are perpendicular and bisect each other.

C. Square

Definition 5.7: square is a parallelogram with four congruent sides and four right angles.

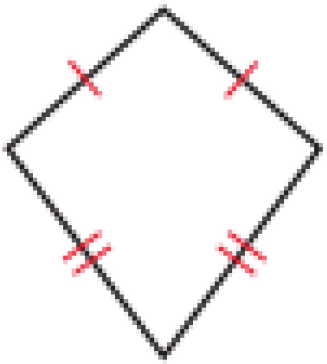


Properties of square:

1. Square satisfies all properties of parallelogram, rectangle and rhombus.
2. The diagonals of square are:
   - perpendicular to each other.
   - congruent
   - bisect each other.
3. The diagonals bisect the angles at the vertices. Hence, the diagonal forms 45^0 with its side.

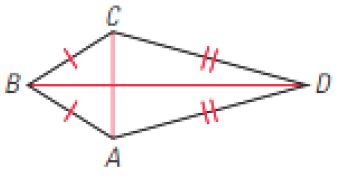
D. Kite

Definition 5.8: Kite is a quadrilateral that has two pairs of consecutive congruent sides, but opposite sides is not congruent.



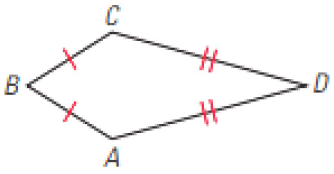
Properties of kite

1. The diagonals of kite are perpendicular to each other, but they do not bisect each other.



$\overline{AC} \perp \overline{BD}$

2. One pair of opposite angles of kite are congruent.



$\angle A \cong \angle C, \angle B \ncong \angle D$

Exercise 5.2.2

1. Write true if the statement is correct and write false if the statement is wrong.
   a. All rectangles are parallelogram.
   b. The diagonals of rhombus are congruent
   c. The opposite sides of kite are congruent
   d. All squares are rhombus
   e. A quadrilateral with congruent diagonals is necessarily rectangle
   f. All trapeziums are quadrilateral
2. Fill in the blank space with the correct answer
   a. A quadrilateral with both pair of opposite sides are parallel is_____.
   b. A quadrilateral with adjacent sides congruent, but opposite sides are not congruent is_____.
   c. A rhombus with congruent diagonals is _____.
   d. A quadrilateral with exactly one pair of opposite sides parallel is_____.
   e. A parallelogram with all four angles are right angle, but the diagonals are not perpendicular is _____.

f. A parallelogram in which its diagonals are congruent, perpendicular and bisect each other is_____.

3. Answer the following questions
   a. Describe similarity and difference of rhombus and kite.
   b. Describe similarity and difference of square and rectangle.

5.3. Perimeters and Areas of triangles

At the end of this section you should be able to:

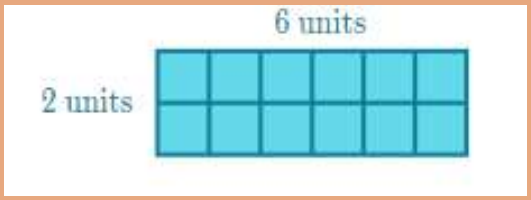
- *solves routine and non-routine problems involving area of composite figures formed by any two or more of the following: triangle, square, and rectangles.*

In your earlier studies you have learnt how to find perimeter and area of rectangle. In this section, after revising about area and perimeter of rectangles you will learn how to find the area and perimeter of triangles.

Activity 5.3.1

Discuss with your friends/partners

1. considering the following rectangle answer the following questions?



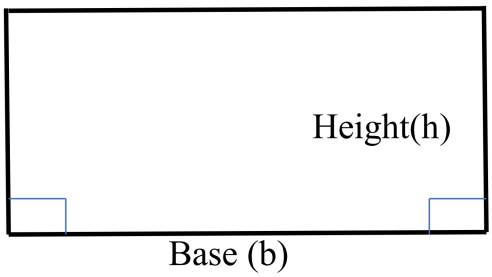
   a. What is the area of the rectangle above in square units?
   b. What is the perimeter of the rectangle?
2. What is area of a closed region?
3. What is perimeter of a closed region?
4. How do you calculate area and perimeter of a rectangle?

Definition 5.9:

Area of a closed figure is the number of square units inside that closed figure.

Perimeter is the length of the boundary of a closed figure.

1. Area and perimeter of rectangle



- Area of rectangle is the product of its base and height.

$$Area\ of\ rectangle = base \times height$$

$$\boxed{A = b \times h}$$

- The perimeter of rectangle is calculated as:

$$perimeter = b + h + b + h$$

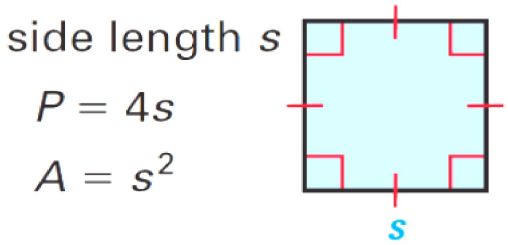
$$P = 2b + 2h$$

$$\boxed{P = 2(b + h)}$$

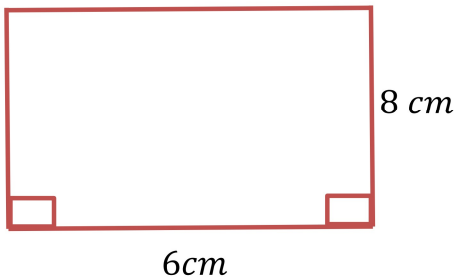
2. Area and perimeter of square

- Square is a rectangle whose base and height are equal.

The area of square whose side length $'S'$ is $A = S^2$ and its perimeter is $P = 4S$.



Example 1: Calculate the area and perimeter of the rectangle given below.



Solution:

$$A = b \times h$$

$$A = 6cm \times 8cm = 48cm^2$$

$$P = 2(b + h)$$

$$P = 2(6 + 8) = 2 \times 14 = 28cm$$

Example 2: calculate the area and perimeter of square whose side length is $10m$

Solution: $S = 10m, A = ?, P = ?$

- $A = S^2$

  $A = (10m)^2 = 100m^2$

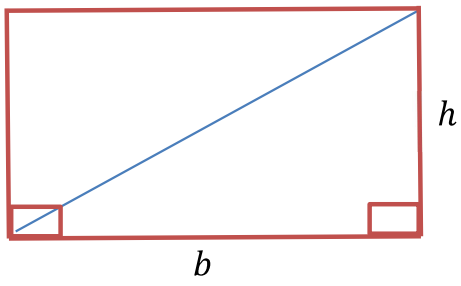
- $P = 4S$

  $P = 4 \times 10m = 40$

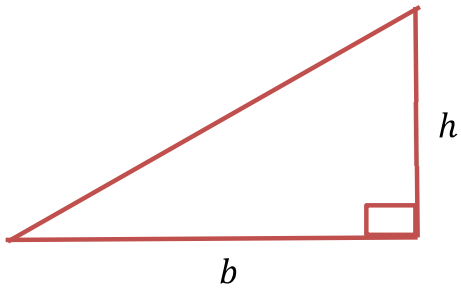
3. Area and perimeter of triangle

a. Area of right angled triangle

The formula of area of right angled triangle is derived from area of rectangle. The diagonal of a rectangle divides the rectangle in to two equal right-angled triangles.



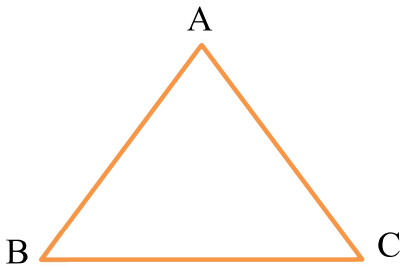
$A = bh$



The area of triangle is half of the area of rectangle , so the area of triangle is:

$$A = \frac{1}{2}bh$$

b. Area of acute angled triangle

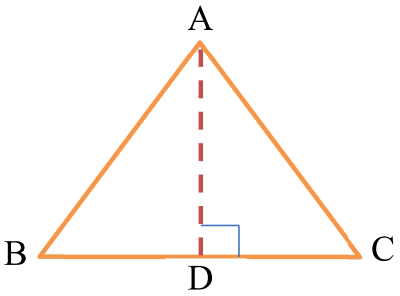


To calculate the area of such triangle, draw perpendicular line from one of the vertices to the opposite side.

The perpendicular line segment from the vertex to the opposite side is called altitude or height(h) and the opposite side is the base(b).

AD is height (h) and BC is base(b). The area of ΔABC is the sum of area of $\Delta ABD \text{ and } \Delta ADC$

Area of $\Delta ABC = area\ of\ \Delta ABD + area\ of \Delta ADC$



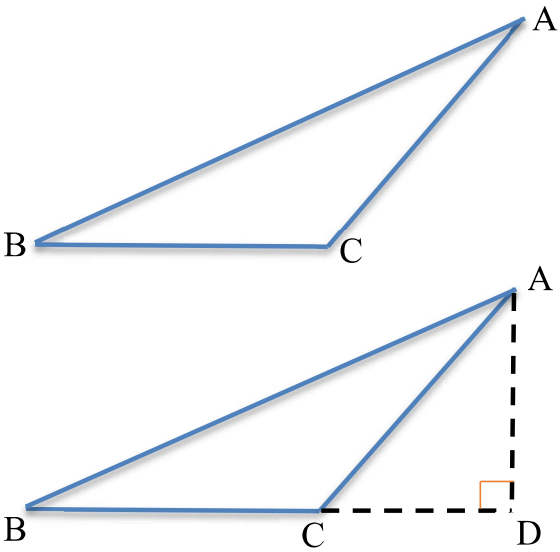
$$A = \frac{1}{2}BD \times AD + \frac{1}{2}AD \times DC$$

$$A = \frac{1}{2}AD \times (BD + DC)$$

$$A = \frac{1}{2}AD \times BC \ldots\ldots\ldots\ldots\ldots (BD + DC = BC)$$

$$A = \frac{1}{2}bh \ldots\ldots\ldots where\ AD\ is\ height(h)\ and\ BC\ is\ base(b)$$

c. *Area of obtuse angled triangle*



- To calculate the area of obtuse angled triangle, draw height from vertex A to side extension of base BC.
- Area of ΔABC is equal to area of ΔABD minus area of ΔADC.

- Area of $\Delta ABC = area\ of\ \Delta ABD - area\ of \Delta ADC$

$$A = \frac{1}{2}BD \times AD - \frac{1}{2}AD \times DC$$

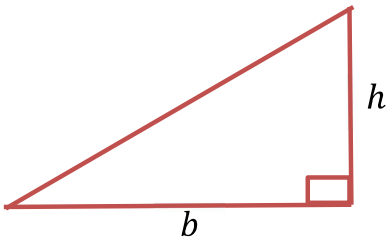
$$A = \frac{1}{2}AD \times (BD - DC)$$

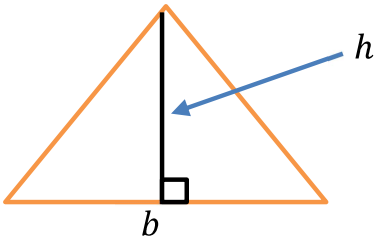
$$A = \frac{1}{2}AD \times BC \ldots\ldots\ldots\ldots\ldots (BD - DC = BC)$$

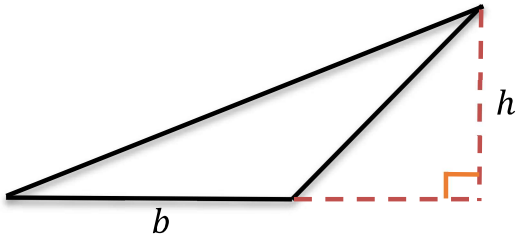
$$A = \frac{1}{2}bh \ldots\ldots\ldots where\ AD\ is\ height(h)\ and\ BC\ is\ base(b)$$

Note:

The area of triangle whose base b and altitude(h) to this base is given by:



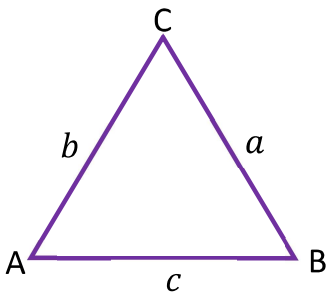




$$A = \frac{1}{2}bh$$

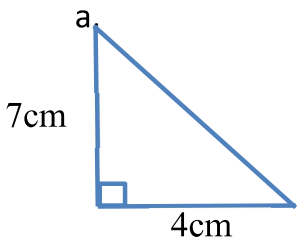
Note: If the lengths of the sides of a triangle are *a, b and c,* then the perimeter *p* of the triangle is:

$$p = a + b + c$$

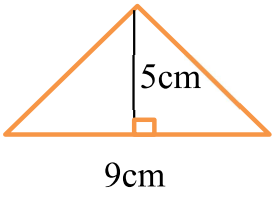


Example 3: Calculate the area of the following triangles.

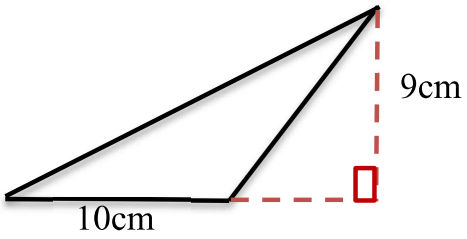
a.



b.



c.



Solution:

a. $A = \frac{1}{2}bh$

$A = \frac{1}{2} \times 7cm \times 4cm$

$A = \frac{28\ cm^2}{2}$

$A = 14cm^2$

b. $A = \frac{1}{2}bh$

$A = \frac{1}{2} \times 9cm \times 5cm$

$A = \frac{45cm^2}{2}$

$A = 22.5cm^2$

c. $A = \frac{1}{2}bh$

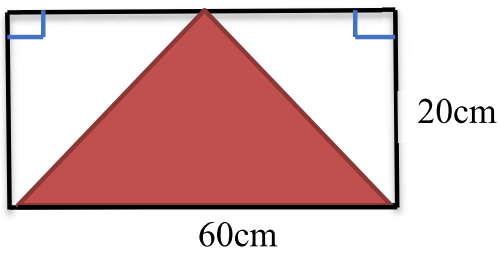
$A = \frac{1}{2} \times 10cm \times 9cm$

$A = \frac{90\ cm^2}{2}$

$A = 45cm^2$

Example 4: Using the figure below, calculate

a. the area of rectangle
b. the area of shaded region
c. the area of unshaded region



Solution:

a. Area of rectangle$= bh$

$= 60cm \times 20cm$

$\underline{\underline{= 1200cm^2}}$

b. Area of shaded part $= \frac{1}{2}bh$

$= \frac{1}{2} \times 60cm \times 20cm$

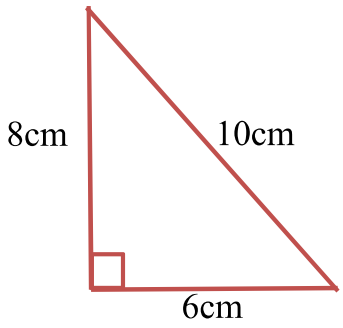
$\underline{\underline{= 600cm^2}}$

c. Area of unshaded part $=$ area of rectangle minus area of shaded part

$= 1200\ cm^2 - 600cm^2$

$\underline{\underline{= 600cm^2}}$

Example 5: what is the area and perimeter of the triangle below



Solution:

- $A = \frac{1}{2} \times 6cm \times 8cm$

  $A = \frac{48\ cm^2}{2}$

  $\underline{\underline{= 24cm^2}}$

- Perimeter of the triangle is sum of each side:

  $P = 8cm + 10cm + 6cm$

Example 6: the area of a triangle is 64 cm^2. If the base is 16 cm long, then calculate the height of the triangle.

Solution: $A = 64cm^2, b = 16cm, h = ?$

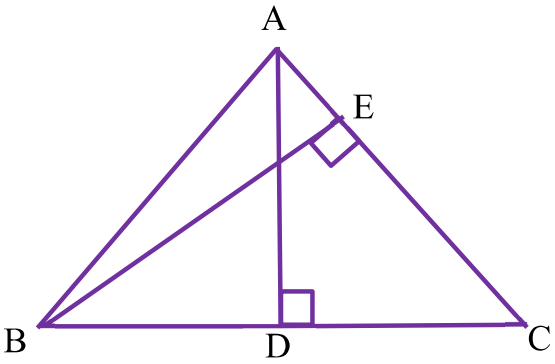
$A = \frac{1}{2}bh$

$64cm^2 = \frac{1}{2} \times 16cm \times h$

$\frac{64cm^2}{8cm} = \frac{8cm \times h}{8cm}$

$\underline{\underline{h = 8cm}}$

Example 7: based on the triangle below answer the following questions.



If $AD = 12cm, BC = 8cm$ *and* $AC = 6cm$, then calculate

a. the area of ΔABC.
b. the length BE.

Solution:

a. $A(\Delta ABC) = \frac{1}{2}bh$

$= \frac{1}{2} \times BC \times AD$ ..... using base BC and height AD

$= \frac{1}{2} \times 8cm \times 12cm$

$\underline{= 48cm^2}$

b. using the other base and height, area of ΔABC is

$A(\Delta ABC) = \frac{1}{2}bh$

$A(\Delta ABC) = \frac{1}{2} \times AC \times BE$ .... using base AC and height BE

$48cm^2 = \frac{1}{2} \times 6cm \times BE$

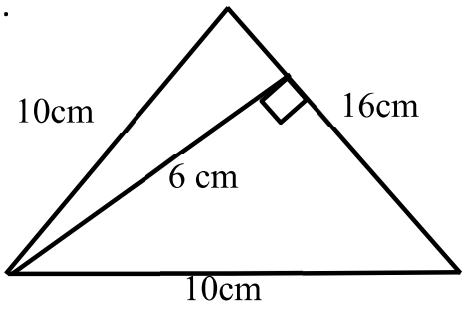
$\frac{48cm^2}{3cm} = \frac{3cm \times BE}{3cm}$

$\underline{BE = 16cm}$

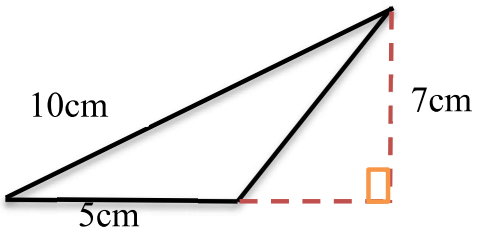
Exercise 5.3.1

1. Calculate the area and perimeter of the following triangles.

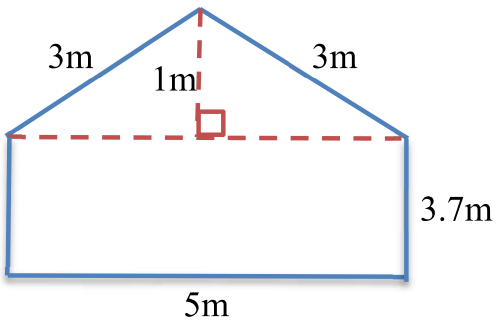
a.



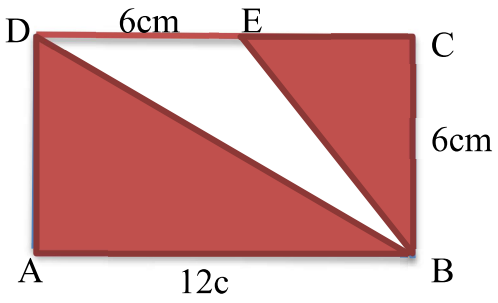
b.



2. The area of a triangle is 63 cm^2. If its height is 9cm long, then calculate the length of its base.
3. The figure below represents a wall of a certain building. Find the area and perimeter of the wall.



4. Based on the figure below, answer the following questions



a. Calculate the area of shaded region

b. Calculate the area of unshaded part

5.4. Perimeter and Area of four-sided figures

At the end of this section you should be able to:

- *solves routine and non-routine problems involving Perimeter and area of four-sided figures.*

In this section you will learn how to calculate, the area and perimeter of parallelogram, trapezium, rhombus and kite.

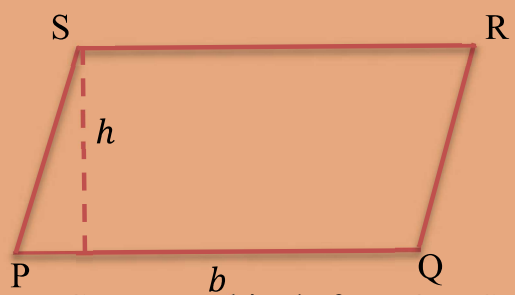
a. Perimeter and area of parallelogram

Activity 5.4.1

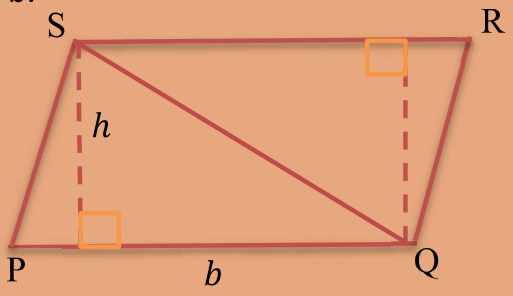
Discuss with your friends/partners

1. What is parallelogram?
2. List its properties of parallelogram.
3. Consider the parallelogram below, where PQ is base(b) and the altitude(h) from S to PQ.

a.



b.



i. Construct altitude from Q to SR, and diagonal SQ as in b above.

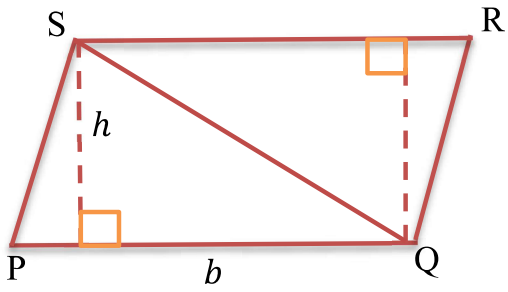
ii. What is the area of ΔPQS in terms of b and h?

iii. What is the area of ΔQRS in terms of b and h?

iv. How do you find the area parallelogram using areas of ΔPQS and ΔQRS?

v. From ii, iii and iv the formula to calculate area of parallelogram is_____.

As you have done in the above activity the formula to calculate area of parallelogram is derived as follows:



$$A(PQRS) = A(\Delta PQS) + A(\Delta QRS)$$

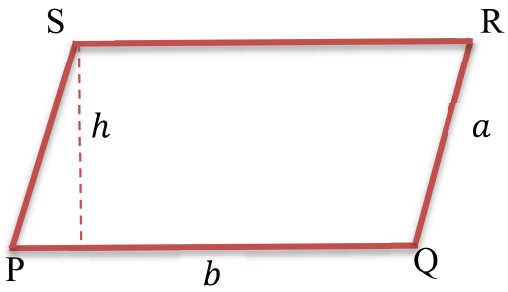
$$A(PQRS) = \frac{1}{2} \times PQ \times h + \frac{1}{2} \times RS \times h$$

$$A(PQRS) = \frac{1}{2}bh + \frac{1}{2}bh$$

$$A(PQRS) = bh$$

Note: 1. The area of parallelogram with base b and altitude h is given by the formula:

$$A = bh$$



2. The perimeter of parallelogram is calculated by adding all its sides

$$P = PQ + QR + RS + PS$$

$$P = b + a + b + a$$

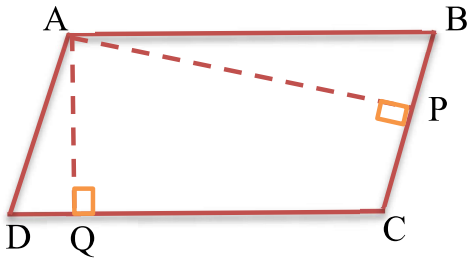
Example 1:

$$P = 2a + 2b$$

$$P = 2(a + b)$$

In the figure AP, AQ, are altitudes of the parallelogram ABCD. If $AQ = 4cm$, $CD = 5cm$ and $AP = 8cm$, then calculate

a. area of the parallelogram
b. length of BC
c. perimeter of the parallelogram



Solution:

a. $A = bh$

$A = DC \times AQ$…. using base DC

And height AQ

$A = 5cm \times 4cm$

$\underline{A = 20cm^2}$

b. $A = bh$

$A = BC \times AP$…. using base B and

$\frac{20cm^2}{8cm} = \frac{BC \times 8cm}{8cm}$

$\underline{BC = 2.5cm}$

c. perimeter of the parallelogram is:

$P = AB + BC + CD + AD$

$P = 5cm + 2.5cm + 5cm + 2.5$, $AB = DC = 2.5cm, AD = BC = 2.5cm$

$\underline{\underline{P = 15cm}}$

Example 2: The area of parallelogram is $42cm^2$.If the base of the parallelogram is $6cm$, then find the corresponding altitude.

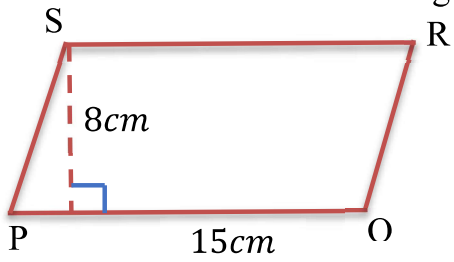
Solution: $A = 42cm^2, b = 6cm, h = ?$

$A = bh$

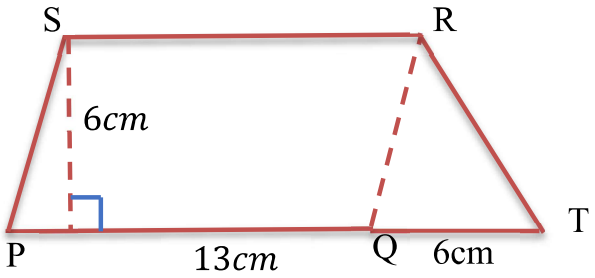
$\frac{42cm^2}{6cm} = \frac{6cm \times h}{6cm}$, $\underline{h = 7cm}$

Exercise 5.4.1

1. Calculate the area of the following parallelogram

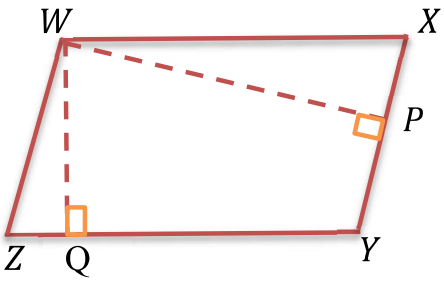


2. ABCD is a parallelogram of area $18cm^2$. Find the length of the corresponding altitudes, if $AB = 5cm$
3. Calculate the area of the region $PTRS$, if PQRS is parallelogram



4.

In the figure WQ, WP, are altitudes of the parallelogram $WXYZ$. If $WP = 6cm$, $XY = 5cm$ and $WQ = 8cm$, then calculate



a. area of the parallelogram.

b. length of ZY.

c. perimeter of the parallelogram.

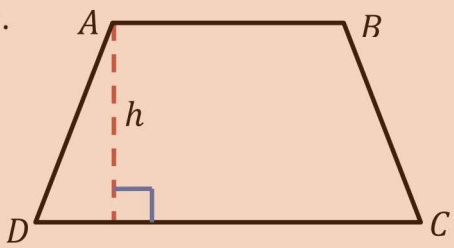
b. Perimeter and area of trapezium

Activity 5.4.2

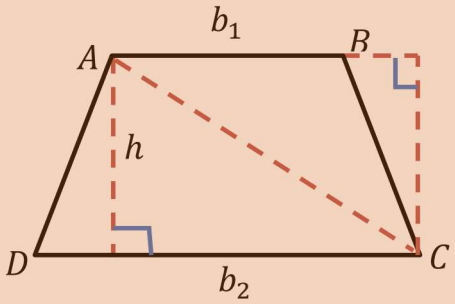
Discuss with your friends/partners

1. What is trapezium?
2. Consider the trapezium below

a.



b.



i. Construct diagonal AC and altitude from c to base AB (its extension) as in b above.

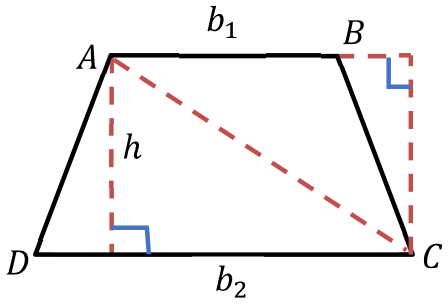
ii. What is the area of ΔADC in terms of b_2 *and* h?

iii. What is the area of ΔABC in terms of b_1 *and* h?

iv. The area of trapezium ABCD is the sum of the area of ΔADC and ΔABC.

v. From ii, iii and iv the formula to calculate area of trapezium is _____.

As you have done in the above activity the formula to calculate area of trapezium is derived as follows:



- The area of trapezium ABCD is the sum of the area of ΔADC and ΔABC

$$A(ABCD) = A(\Delta ADC) + A(\Delta ABC)$$

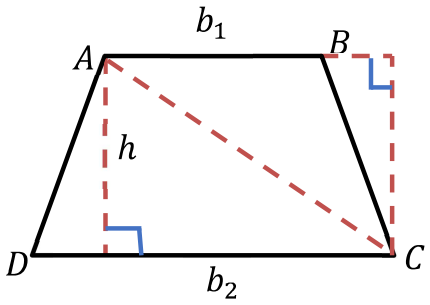
$$= \frac{1}{2} \times b_2 \times h + \frac{1}{2} \times b_1 \times h$$

$$= \frac{1}{2} h(b_1 + b_2)$$

$$A = \frac{h}{2}(b_1 + b_2)$$

Note: 1. Area of trapezium with base b_1 and b_2 and altitude h is given by the formula:

$$A = \frac{h}{2}(b_1 + b_2)$$



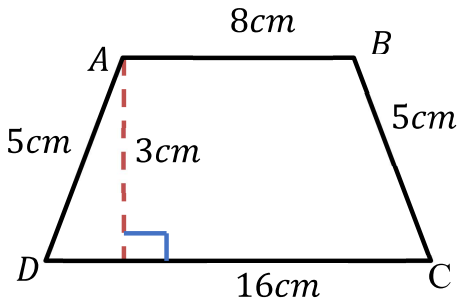
2. The perimeter of trapezium is the sum of the two base and its legs:

$$perimeter = AB + BC + DC + AD$$

$$P = b_1 + b_2 + l_1$$

Example 3: Calculate the area and perimeter of the trapezium given below

Solution:



$$A = \frac{h}{2}(b_1 + b_2)$$

$$A = \frac{3}{2}(8 + 16)$$

$$A = \frac{3}{2} \times 24$$

$$\underline{A = 36cm^2}$$

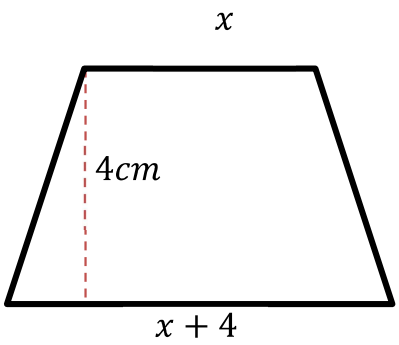
$$P = b_1 + b_2 + l_1 + l_2$$

$$P = 8 + 16 + 5 + 5$$

$$\underline{P = 34cm}$$

Example 4: The area of trapezium is $48cm^2$ and its height is $4cm$. If one of the bases is $4cm$ more than the other base, then calculate each base of the trapezium.

Solution: let the smaller base is x, and the larger base is $x + 4$



$$A = 48cm^2$$

$$A = \frac{h}{2}(b_1 + b_2)$$

$$48 = \frac{4}{2}(x + x + 4)$$

$$48 = 2(2x + 4)$$

$$4x + 8 = 48$$

$$4x = 40$$

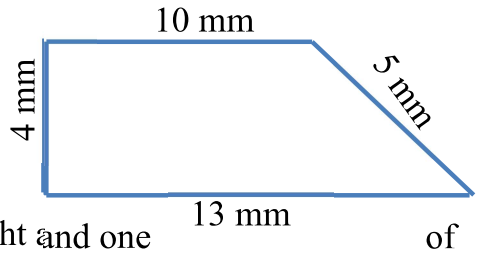
Therefore, the bases of trapezium are:

Smaller base $x = 10cm$ and

larger base is $x + 4 = 10 + 4 = 14cm$.

Exercise 5.4.2

1. Calculate the area and perimeter of the following trapezium.



2. The area of trapezium is $170cm^2$. If its height and one of the bases are 17cm and 12 cm respectively, then calculate the other base of the trapezium.
3. One of the bases of trapezium exceeds the other by $2cm$. If the altitude and area of trapezium are $6cm$ and 42 cm^2 respectively, then calculate the larger base of the trapezium.

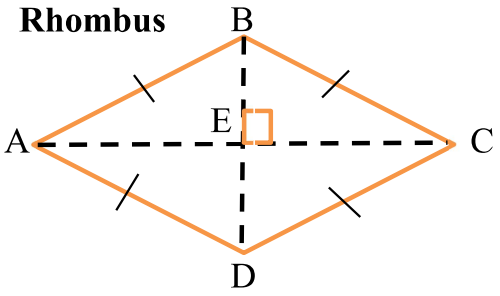
c. Perimeter and area of rhombus and kite

Activity 5.4.3

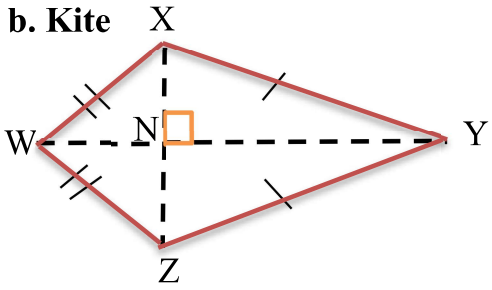
Discuss with your friends/partners

1. Define rhombus and list its properties.
2. Define kite and list its properties
3. Consider the following rhombus and kite

a. Rhombus



b. Kite



i. Name the diagonals of the rhombus.

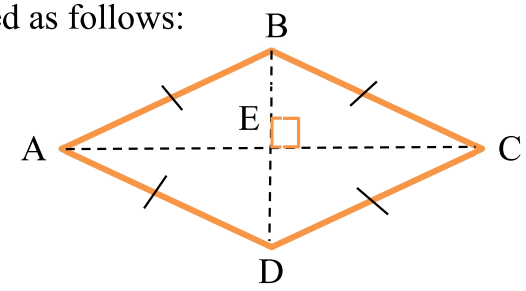
ii. Name the diagonals of kite.

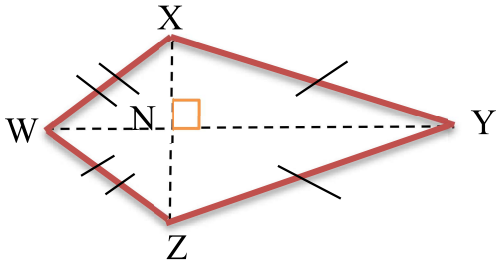
iii. Calculate the area of rhombus by adding area of ΔABC *and* ΔADC.

iv. Calculate the area of kite by adding area of ΔWXY *and* ΔWZY.

v. Using the above result generalize the formula to calculate area of rhombus and kite.

As you have done in the above activity the formula to calculate area of trapezium is derived as follows:





Area of rhombus is the sum of area ΔABC *and* ΔADC

$A = A(\Delta ABC) + A(\Delta ADC)$

$A = \frac{1}{2}AC \times BE + \frac{1}{2}AC \times ED$

$A = \frac{1}{2}AC(BE + ED)$

$A = \frac{1}{2}AC \times BD$ … since $BE + ED = BD$

$A = \frac{1}{2}d_1d_2$, where AC is diagonal $1(d_1)$ and BD is diagonal $2(d_2)$

Area of kite is the sum of area ΔWXY *and* ΔWZY

$A = \Delta WXY + \Delta WZY$

$A = \frac{1}{2}WY \times NX + \frac{1}{2}WY \times NZ$

$A = \frac{1}{2}WY(NX + NZ)$

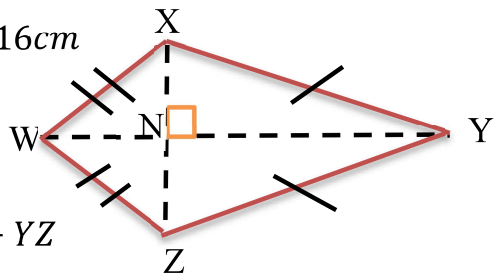
$A = \frac{1}{2}WY \times XZ$, since $NX + NZ = XZ$

$A = \frac{1}{2}d_1d_2$, where WY is diagonal$1(d_1)$ and XZ is diagonal $2(d_2)$

Note: The area of rhombus and kite is given by the formula $A = \frac{1}{2}d_1d_2$, where d_1 and d_2 are diagonals. The perimeter of rhombus and kite is calculated by adding the length of all the four sides.

Example 5: Calculate the perimeter and area of the kite given below,

if $WX = 6cm, XY = 10cm, XZ = 8cm$ *and* $WY = 16cm$



Solution:

$A = \frac{1}{2}d_1d_2$

$A = \frac{1}{2}XZ \times WY$

$A = \frac{1}{2} \times 8 \times 16$

$\underline{A = 64\ cm^2}$

$P = WX + WZ + XY + YZ$

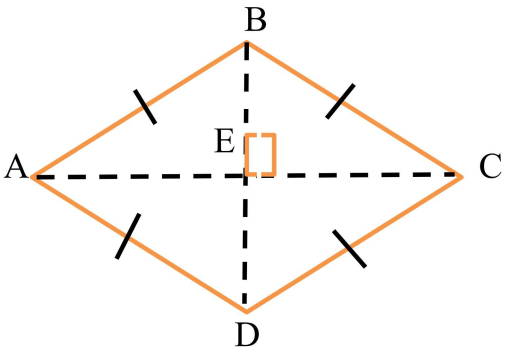
$P = 6 + 6 + 8 + 8$

$\underline{P = 18cm}$

Exercise 5.4.3

1. Calculate the area and perimeter of the following rhombus if

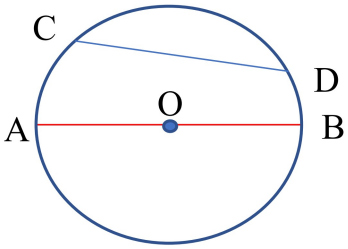
   $AB = 5cm, AC = 8cm$ *and* $BD = 6cm.$



2. Calculate the area of kite, whose diagonals are 12cm and 16cm.
3. Calculate the perimeter of kite, whose adjacent sides are 18mm and 10mm.
4. The area of rhombus is $144cm^2$. If one of its diagonals is 18cm long, then calculate the length of the other diagonal.

5.5. Circumference and Area of a circle

Definition 5.10: A circle is the set of all points in a plane that are equidistant from a fixed point called the center of the circle.



Circle

Note:

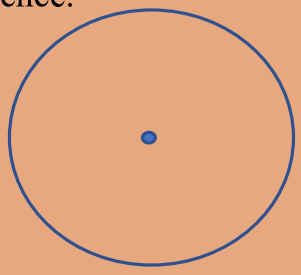
1. a circle is usually named by its center. The above circle is named as circle O.
2. a chord is a line segment whose end points are on the circle. In the circle above $\overline{CD}$ and $\bar{A}$ are chords of the circle.
3. a diameter is any chord that passes through the center. The chord $\overline{AB}$ is the diameter of the circle. Diameter is the longest chord of a circle. It is denoted by 'd'.
4. a radius is a line segment from the center to any point on the circle. It is denoted by 'r'. The plural form of radius is radii.
5. Diameter is twice of the radius (i.e. $d = 2r$).
6. Circumference is the complete path around the circle. It is the perimeter of a circle and denoted by 'C'.

Formula for circumference of a circle

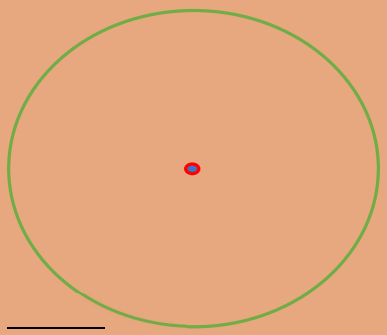
Activity 5.5.1

Discuss with your friends:

1. Put a thread slowly and carefully around each of the circle until its both ends join (do not overlap the threads).
2. Stretch this thread against a ruler and measure its length which gives the circumference.



$C =$ _____
$d =$ _____
$\frac{C}{d} =$ _____



$C =$ _____
$d =$ _____
$\frac{C}{d} =$ _____

3. From above activity, what do you conclude for the ratio $\frac{C}{d}$?

   Note: From the above activity you can notice that the ratio of circumference to diameter of a circle, whether the circle is big or small is constant.

   This constant value is approximately is equal to 3.14 or $\frac{22}{7}$ or symbolically π(read as pi).

   So, $\frac{C}{d} = \pi \Rightarrow C = \pi \times d$.

   The circumference of a circle whose diameter d is given by:

   $C = \pi \times d$ Or $C = \pi \times 2r$ where c is circumference, d is diameter and r is radius.

Example 1: Find the circumference of a circle with diameter 6cm (use $\pi = 3.14$).

Solution: $C = \pi \times d$

$$C = \pi \times 6cm = 3.14 \times 6cm = 18.84cm$$

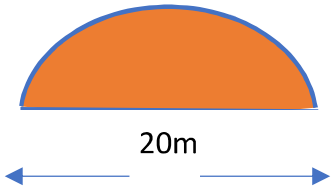
Example 2: Find the circumference of a circle with radius 7cm (use $\pi = \frac{22}{7}$).

Solution: $C = \pi \times 2r$

$$C = \pi \times 2 \times 7cm = \frac{22}{7} \times 2 \times 7cm = 44cm$$

Exercise 5.5.1

1. Find the circumference of the circle with each of the given diameters below (use $\pi = 3.14$).
   a. 4m b. 10cm
2. Find the circumference of the circles with radii is given below (use $\pi = \frac{22}{7}$).
   a. 14cm b. 49m
3. A piece of land has a shape of semicircular region as shown below.

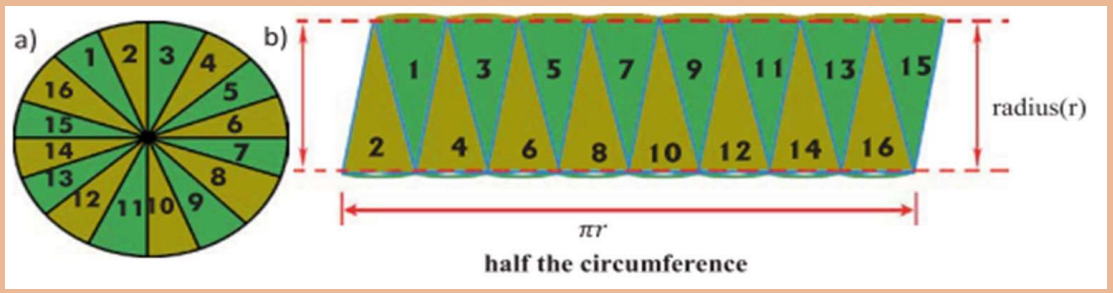


Find the perimeter of the land (use $\pi = 3.14$).

Formula for area of a circle

Activity 5.5.2

1. Draw a circle of radius 10 cm and divide the circle in to 16 equal parts and try to fill a rectangle as in the figure below.



2. The length of rectangle is equal to_____
3. The width of rectangle is equal to _____
4. The area of rectangle which is equal to area of circle is _____.

From above activity,

The area of rectangle =length ×width

$$= \pi r \times r$$

$$= \pi r^2$$

This is the same as the area of the circle, so

Area of a circle (A) $= \pi r^2$

The area of a circle whose radius r unit long is given by:

A= πr^2 Or A= $\pi r^2 = \pi(\frac{d}{2})^2 = \frac{\pi d^2}{4}$.... since $r = \frac{d}{2}$

Example 3: Find the area of a circle with radius 10cm (use $\pi = 3.14$).

Solution: A= πr^2

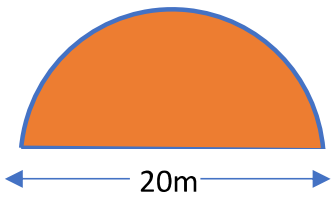
$$A = \pi \times (10cm)^2 = 3.14 \times 100cm^2 = 314cm^2$$

Example 4: Find the area of a circle with diameter 8m. (leave your answer in terms of π).

Solution: A = $\frac{\pi d^2}{4}$

$$= \frac{\pi(8m)^2}{4} = \frac{\pi \times 64m^2}{4} = 16\pi m^2$$

Example 5: Find the area of the semicircle given below



Solution: $d = 20m, so\ r = 10m$

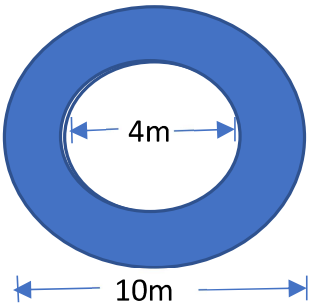
- The area of semi-circle is equal to half of the area of circle, so;

$$A = \frac{1}{2} \times \pi r^2$$

$$A = \frac{1}{2} \times \pi(10m)^2 = \frac{100\pi m^2}{2} = 50\pi m^2$$

Exercise 5.5.2

1. Find the area of a circle with each of the given radii below.
   (use $\pi = 3.14$).
   a. 4m b. 100cm
2. Find the area of the circles with each of the given diameters below
   (leave your answer in terms of π).
   a. 10cm b. 16m
3. Find the area of the shaded region below (use $\pi = 3.14$).



5.6. Applications

At the end of this section you should be able to:

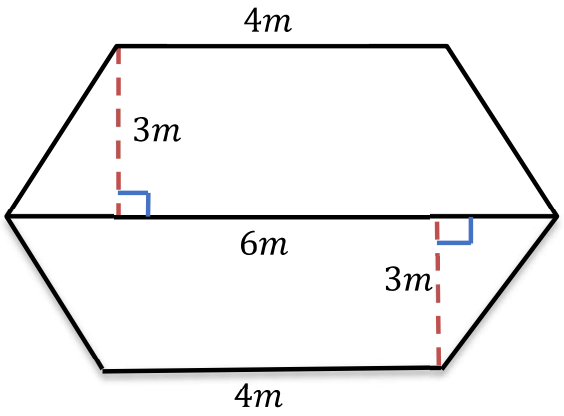
- *Apply the concept of perimeter and area of plane figures in real life problems.*

In this section you will learn some of the application of area and perimeter in our real world.

The concept of area and perimeter is applied:

- In decorating the wall of a house, so that to estimate the cost of painting.
- To estimate the fencing material required in fencing a garden.
- To estimate the area of roofing material.
- Making various furniture.
- To estimate carpet size for floor.

Example 1: the diagram below shows the floor plan of a hotel. Carpet costs Birr 100 per square meter. How much will it costs to carpet the hotel.



Solution: First calculate the area of the floor

Area of the floor is the sum of the areas of the two trapeziums.

$$A = \frac{h}{2}(b_1 + b_2) + \frac{h}{2}(b_1 + b_2)$$

$$A = \frac{3}{2}(4 + 6) + \frac{3}{2}(4 + 6)$$

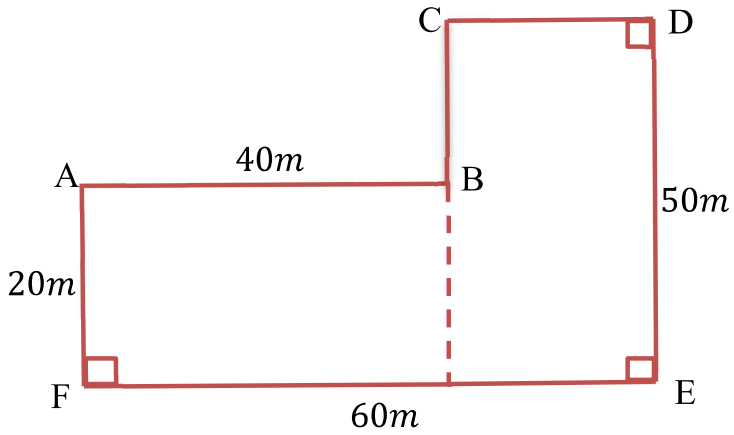
$$A = 15 + 15$$

$$= 30m^2$$

The cost of $1m^2$ carpet is Birr 100, so the cost of $30m^2$ is $Birr\ 30 \times 100 = 3000$

Hence, it costs Birr 3000 to carpet the floor.

Example 3: A farmer wants to fence a plot of land shown below.



a. Calculate the length of fencing material required.

b. If the cost of 1m fencing material is Birr 250, then calculate the cost to fence the plot of land.

Solution:

a. $P = AB + BC + CD + DE + EF + FA$

$P = 40 + 30 + 20 + 50 + 60 + 20$

$P = 220m$

Therefore, 220m of fencing material is required.

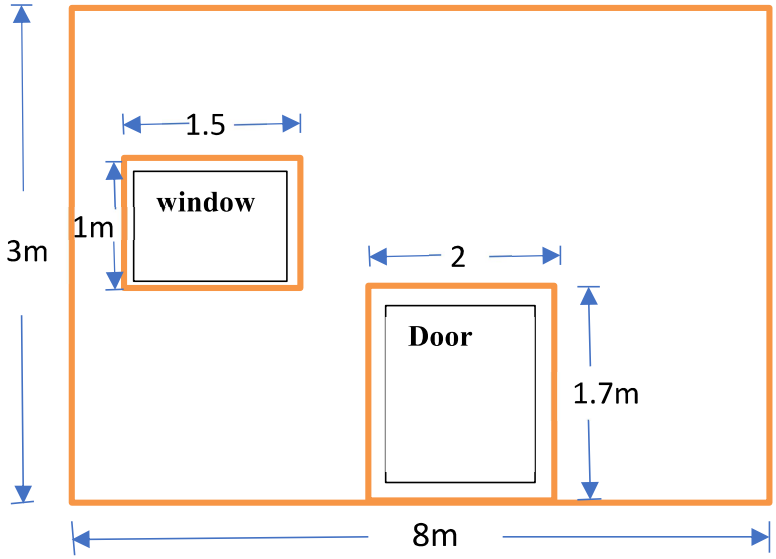
b. The cost of 1m fencing material is Birr 250, so the cost of 220m fencing material is

$Birr250 \times 220 = Birr\ 55{,}000$

- $BC = DE - AF = 50 - 20 = 30$

and $CD = EF - AB = 60 - 40 = 20$

Example 3: The following diagram shows the front part of a building



If you paint the wall of the building, then,

a. calculate the area of the building painted.
b. If it costs Birr200 to paint $1\ m^2$ wall, then how much it costs to paint the wall?

Solution:

a. Note that the area of window and door will not be painted, so the area of wall painted is equal to the area of bigger rectangle minus the area of door and window

$$painted\ area = 8 \times 3 - 1 \times 1.5 - 2 \times 1.7$$
$$= 24 - 1.5 - 3.4$$
$$\underline{\underline{= 19.1m^2}}$$

b. it costs Birr200 to paint $1\ m^2$. So, to paint $19.1m^2$, it costs $Birr\ 200 \times 19.1 = Birr\ 3820$

Example 4: A class room has length of 9m and width of 6m. The flooring is to be replaced by terazo tiles of size 30cm by 30cm. how many terazo tiles are needed to cover the class room?

Solution: Area of class room, $A = 9m \times 6m = 54m^2$

$$= 540{,}000cm^2$$

$1\ m^2 = 10{,}000\ cm^2$

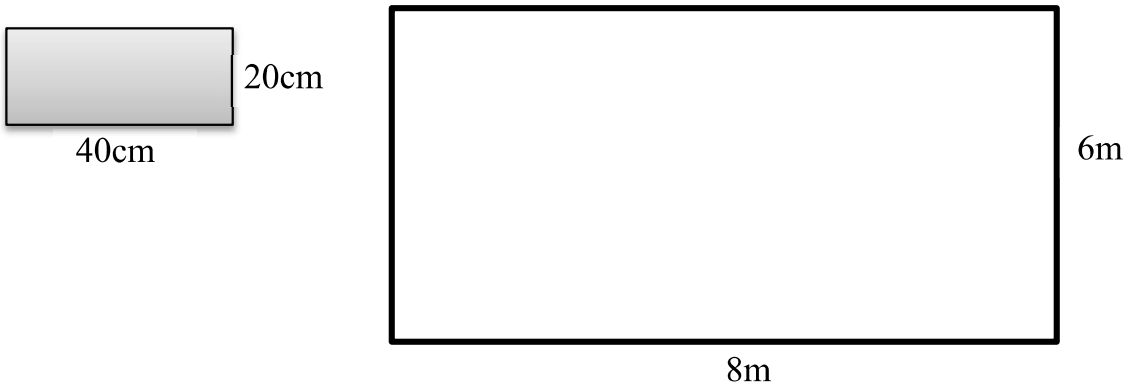
Area of terazo tile, $A = 30cm \times 30cm = 900cm^2$

$$\text{Number of terazo tiles} = \frac{\text{Area of class room}}{\text{Area of terazo tile}} = \frac{540{,}000cm^2}{900cm^2} = 600$$

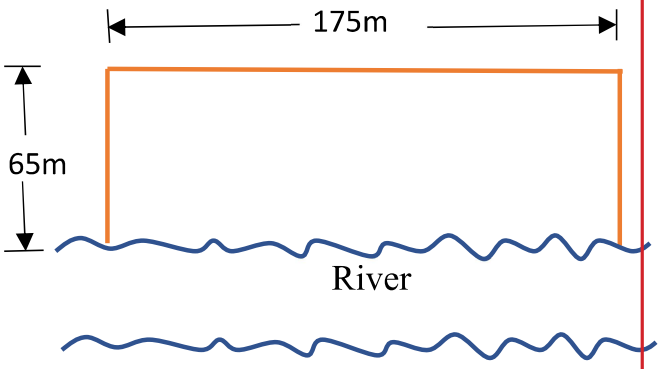
Therefore, $\underline{\underline{600}}$ terazo tiles are required for flooring the class room.

Exercise 5.6.1

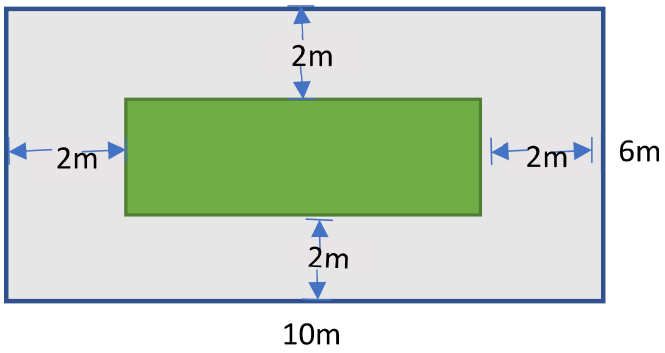
1. The length and width of rectangular football field are 100m and 80m respectively. 1 m^2 artificial grass costs Birr 500, then how much it costs to cover the field by artificial grass.
2. The following diagram shows one side of a building. If this part will be covered by HCB (hollow concrete block), then how many HCB is required? (the size of HCB is 20cm by 40cm)



3. A farmer wants to fence the following plot of land. If no fencing material is not required along river side, then calculate



   a. The length of fencing material required
   b. If the cost of 1 m fencing material is Birr 150, then calculate the cost to fence the land.
4. A rectangular flower bed measures 10m by 6m. It has a path 2m wide around it. Find the area of the path.



Summary for unit 5

- Triangles based on their sides and angles classified as follows

Based on sides	Based on angles
1. Scalene Triangle 2. Isosceles Triangle 3. Equilateral Triangle	1. Acute Angled Triangle 2. Right-Angled Triangle 3. Obtuse Angled Triangle

- A quadrilateral is a four-sided geometric figure bounded by line segments.
- A trapezium is a special type of quadrilateral in which exactly one pair of opposite sides are parallel.
- A parallelogram is a quadrilateral in which both pair of opposite sides are parallel.
- A rectangle is a parallelogram with all its angles are right angle.
- A square is a parallelogram with four congruent sides and four right angles.
- A rhombus is a parallelogram in which all its sides are congruent.
- A Kite is a quadrilateral that has two pairs of consecutive congruent sides, but opposite sides are not congruent.
- Formula to calculate area and perimeter of different geometric figures

1. Triangle $A = \frac{1}{2}bh$, and $P = a + b + h$ 2. Rectangle $A = bh$, and $P = (b + h)$ 3. Square $A = S^2$ and $P = 4S$	4. Rhombus $A = \frac{1}{2}d_1d_2$, $P = 4S$ 5. kite $A = \frac{1}{2}d_1d_2$, $P = 2(a + b)$ 6. trapezium $A = \frac{h}{2}(d_1 + d_2)$ $P = a + b + b_1 + b_2$ 7. circle A= πr^2 and $C = \pi \times d$

Review exercise for unit 5

I. Write True for the correct statements and False for the incorrect statements.

1. If the adjacent sides of a parallelogram are congruent, then the parallelogram is rhombus
2. If the adjacent sides of a parallelogram are congruent, then the parallelogram is square.
3. The diagonals of rectangle make 45^0 with its sides.
4. The diagonals of kite bisect each other.
5. If the diagonals of a parallelogram are perpendicular bisector of each other, then the parallelogram is necessarily rhombus.
6. All squares are rhombus.
7. All rectangles are squares.

II. Fill in the blank space with the correct answer

1. A triangle with all its sides are equal is called_____.
2. A triangle with one right angle is called_____.
3. A quadrilateral with adjacent sides congruent, but opposite sides are not congruent is called_____.
4. A parallelogram with all four sides congruent is necessarily _____.
5. A rhombus with four right angles is _____.

III. Choose the correct answer

1. The area of triangle with base 4cm and height 3cm is _____

   A. $12cm^2$ B. $6cm^2$ C. $7cm^2$ D. $14cm^2$

2. The area of rhombus whose diagonals are 12cm and 18cm is____

   A. $108cm^2$ B. $216cm^2$ C. $80cm^2$ D. $160cm^2$

3. A toilet room has length of 2m and width of 2m. The flooring is to be tiled by ceramic tiles of size 20cm by 20cm. how many ceramic tiles are needed to cover the toilet room?

A. 300 B. 150 C. 200 D. 100

4. The area of rectangular field is $480m^2$. If the length of base of the field is 12m, then the perimeter of the field is _____

A. $40m$ B. $52m$ C. $104m$ D. $80m$

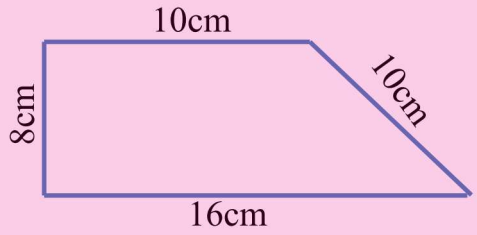
5. The perimeter of kite with its opposite sides are 5cm and 12cm is _____

A. $60cm$ B. $34cm$ C. $17m$ D. $30cm$

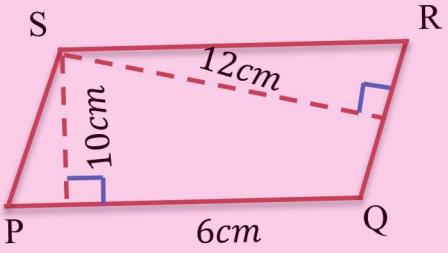
IV. Work out

1. Calculate the area and perimeter of the following figures.

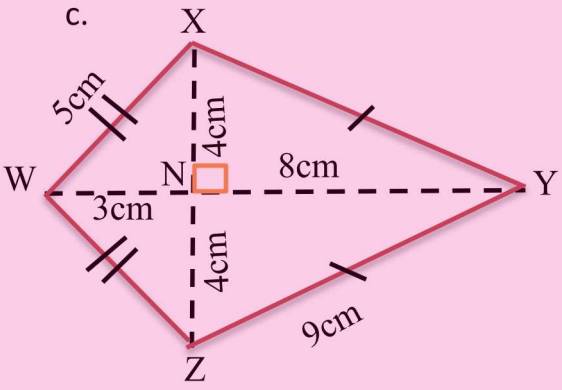
a.



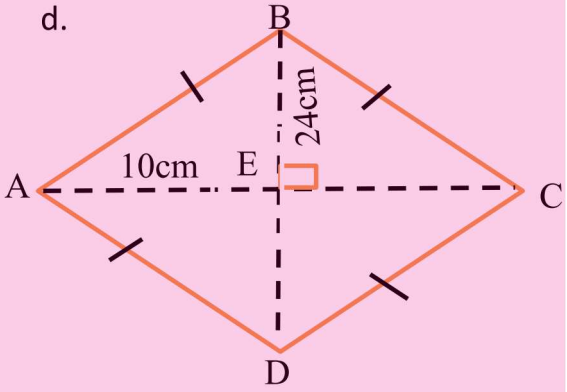
b.



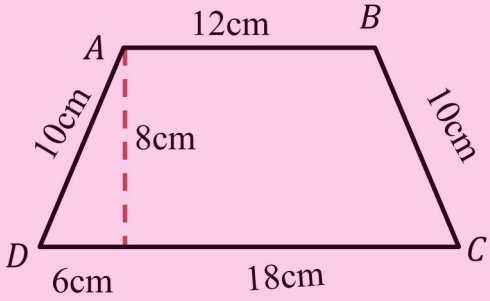
c.



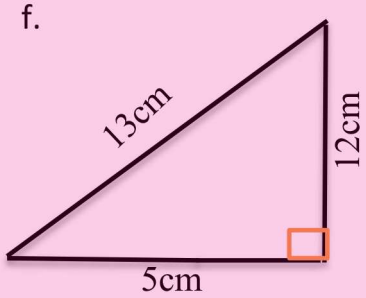
d.



e.



f.



Unit 6
Congruency of plane figures

Learning Outcomes:

At the end of this unit, you will able to:

- *Identify congruent triangles by using the tests for congruency (SSS, SAS, ASA).*
- *Apply real-life situations in solving geometric problems*

Introduction

In this unit you will learn about congruence of geometric figures. The unit has two sections. The first section deals with the congruence of geometric shapes. The main focus of this section is to enable you identifying a given triangle is congruent to the other using congruence test for triangles. In the second section you learn the application of congruence of figures in your daily life.

6.1. Congruent of Plane Figures

At the end of this section you should be able to:

- *explain the concept of congruency of triangles.*
- *check the congruence of given triangles by tracing, cutting and overlapping.*
- *apply real-life applications of congruency of triangles.*

6.1.1 Definition and Illustration of Congruent Figures

Activity 6.1.1

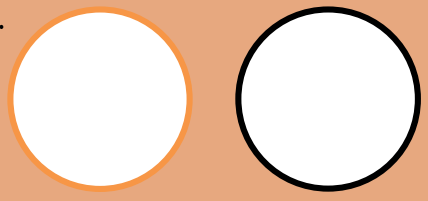
Discuss with your friends/partners

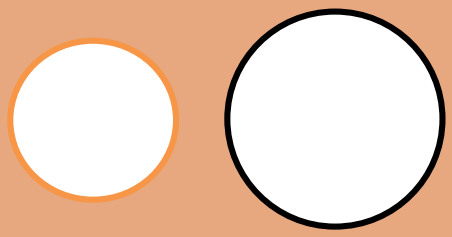
1. Which pair of the following line segments are congruent?

a.

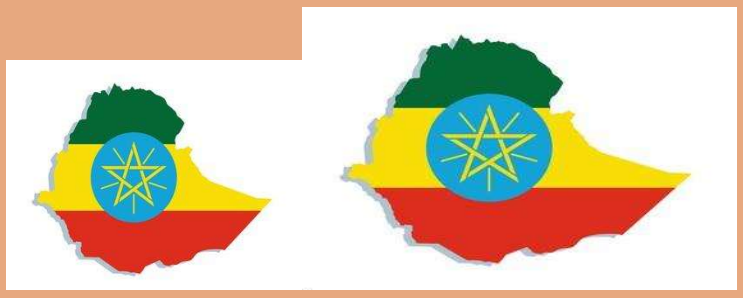
b.

2. Which pair of the following circles are congruent?

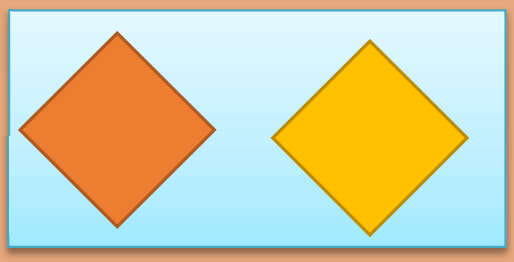
a. 

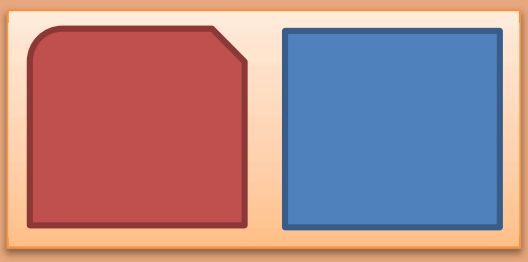
b. 

3. Are the two maps below congruent?



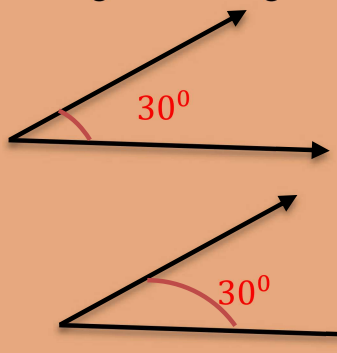
4. Which pair of the following figures are congruent?

a 

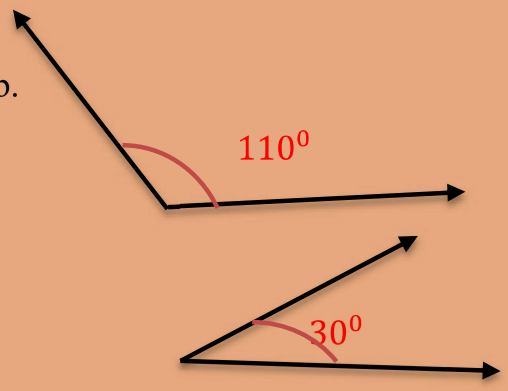
b 

5. Which pair of angles are congruent?

a.



b.



6. When do we say that two figures are congruent?
7. When do we say that two-line segments are congruent?
8. When do we say that two circles are congruent?
9. When do we say that two angles are congruent?

Definition 6.1: Congruent figures are figures that have the same size and shape. Congruent figures are exact copies of one another.

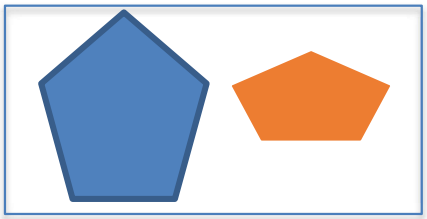
Note:

1. Two figures are congruent, if one figure cover the other completely and exactly.
2. Two-line segments are congruent if they have the same length.
3. Two circles are congruent if they have the same radius.
4. Two angles are congruent if they have the same measure.

Exercise 6.1.1

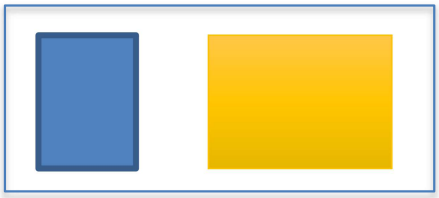
1. Which pair of the following figures are congruent?

a.

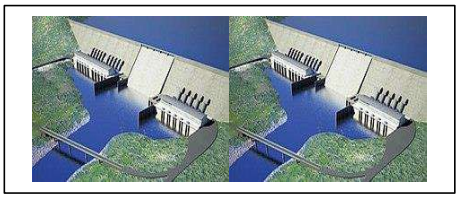


c.

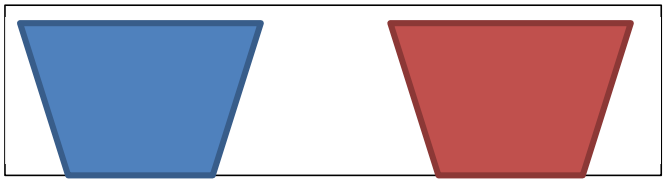
b.



d.



2. Write true if the statement is correct and write false if the statement is wrong.
   a. All line segments are congruent.
   b. All circles are congruent.
   c. All rectangles are congruent.
   d. Two circles having the same diameter are congruent.
   e. Two-line segments having the same length are congruent.
   f. Two angles that have the same measure are congruent.
   g. If two figures have the same shape, then they are congruent.
   h. If two triangles are congruent, then they have the same area.
3. Answer the following questions using the following figures



   a. If you put one of the figures above over the other, does it cover completely and exactly?
   b. What can you say about the figures?

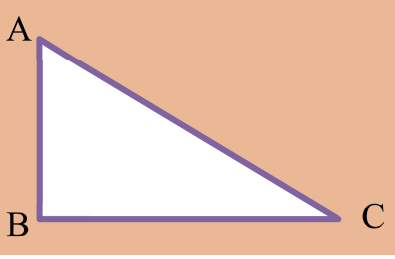
6.1.2 Congruency of Triangles

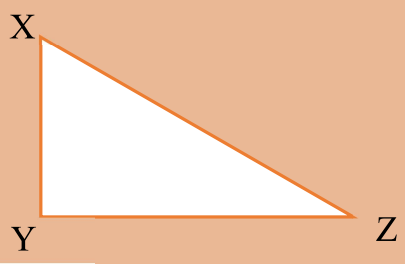
Activity 6.1.2

Discuss with your friends/partners

1. Consider the following two triangles and measure each side and each angle of the two triangles and fill the following blank space

a. Length of AB congruent to ____.
b. Length of BC congruent to ____.
c. Length of AC congruent to ____.
d. $\angle A$ is congruent to _____.
e. $\angle B$ is congruent to _____.
f. $\angle C$ is congruent to _____.





2. If you put ΔABC on ΔXYZ, in such a way that
   a. AB on XY, BC on YZ and AC on XZ, then does ΔABC covers completely ΔXYZ?
   b. What can you say about the two triangles?
3. If ΔABC is congruent to ΔXYZ, then

a. The corresponding side to AB (the side that fit side AB completely) is_____.
b. The corresponding side to AC (the side that fit side AC completely) is _____.
c. The corresponding side to BC (the side that fit side BC completely) is _____.
d. The corresponding angle to $\angle A$ (the angle that fit $\angle A$) is ____.
e. The corresponding angle to $\angle B$ (the angle that fit $\angle B$) is ____.
f. The corresponding angle to $\angle C$ (the angle that fit $\angle C$) is ____.

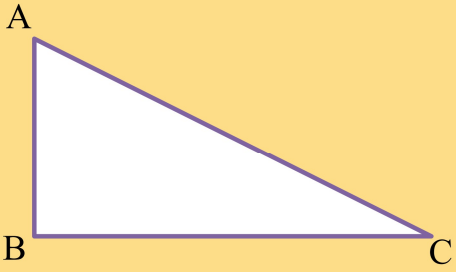
4. When do we say that two triangles are congruent?

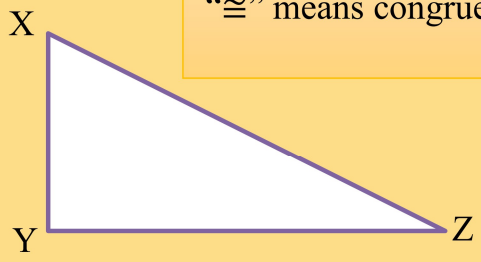
Definition 6.2: Two triangles are congruent, if their corresponding parts (angles and sides) that match one another are equal.

Note:

1. Two triangles are congruent if they are copies of each other and when you place one triangle on another, they cover each other completely.
2. If ΔABC is congruent to ΔXYZ, then symbolically written as $\Delta ABC \cong \Delta XYZ$
3. If $\Delta ABC \cong \Delta XYZ$, then when you place ΔXYZ on ΔABC, it should satisfy the following six conditions.

"≅" means congruent to.





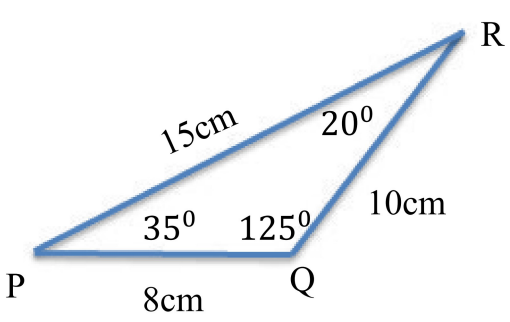
i. X falls on A, and $\angle X \cong \angle A$.
ii. Y falls on B, and $\angle Y \cong \angle B$.
iii. Z falls on C, and $\angle Z \cong \angle C$.
iv. $\overline{XY}$ falls along $\overline{AB}$, and $\overline{XY} \cong \overline{AB}$
v. $\overline{YZ}$ falls along $\overline{BC}$, and $\overline{YZ} \cong \overline{BC}$.
vi. $\overline{XZ}$ falls along $\overline{AC}$, and $\overline{XZ} \cong \overline{AC}$.

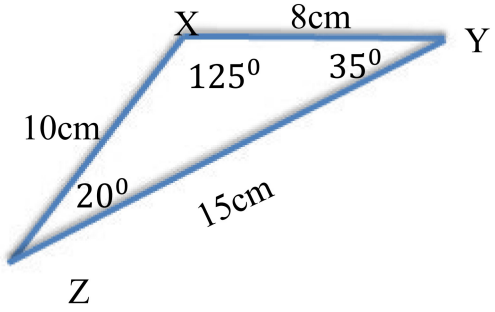
Example 1: If $\Delta ABC \cong \Delta FED$, then find the six congruent corresponding parts of the triangles.

Solution:

i. $\angle A \cong \angle F$
ii. $\angle B \cong \angle E$
iii. $\angle C \cong \angle D$
iv. $\overline{AB} \cong \overline{FE}$
v. $\overline{BC} \cong \overline{ED}$
vi. $\overline{AC} \cong \overline{FD}$

Example 2: consider the following triangles, congruent angles are marked as indicated, based on the figure answer the following questions





a. $\Delta PQR \cong$ _____ b. $\Delta QRP \cong$ _____

Solution:

a. The corresponding angle of $\angle P$ is $\angle Y$, The corresponding angle of $\angle P$ is $\angle Y$.
The corresponding angle of $\angle Q$ is $\angle X$ The corresponding angle of $\angle P$ is $\angle Y$.
And the corresponding angle of $\angle R$ is $\angle Z$, The corresponding angle of $\angle P$ is $\angle Y$.
Therefore, $\Delta PQR \cong \Delta ZYX$.

b. $\Delta QRP \cong \Delta YXZ$.

Example 3: Let $\Delta ABC \cong \Delta DEF$, and $m(\angle A) = 70^0, DE = 6cm$, then find $m(\angle D)$ and the length of AB.

Solution:

Since the triangles are congruent, their corresponding part are congruent.

So, $m(\angle A) = m(\angle D) = 70^0$ and , $DE = \text{AB} = 6cm$

Exercise 6.1.2

1. Write true if the statement is correct and write false if the statement is wrong
   a. If $\Delta ABC \cong \Delta EDF$, then $m(\angle A) = m(\angle D)$.
   b. All equilateral triangles are congruent.

c. If $\Delta ABC \cong \Delta EDF$, then $AB = ED$.

d. If $\Delta ABC \cong \Delta EDF$, then $\angle B \cong \angle D$

e. If $\Delta ABC \cong \Delta EDF$, then $\overline{AC} \cong \overline{DF}$

f. Any triangle is congruent to itself.

2. If $\Delta ZXY \cong \Delta EDF$, then

a. $\Delta XYZ \cong$ _____

b. $m(\angle D) =$ _____

c. $\overline{FD} \cong$ _____

d. $FD =$ ____

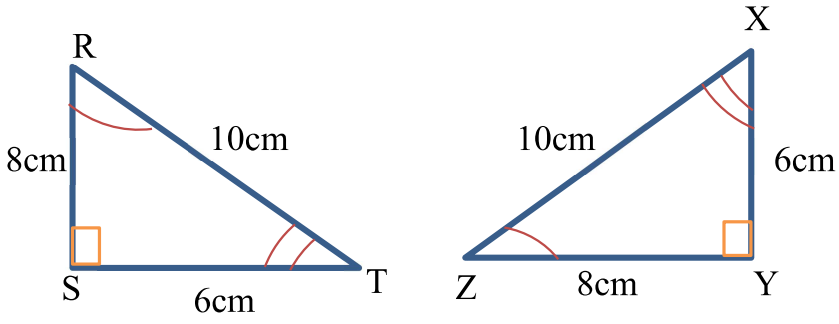
e. $\angle D \cong$ ____

f. $\Delta YXZ \cong$ ____

Note:

1. AB means measure of length of $\overline{AB}$
2. $m(\angle D)$ means measure of $\angle D$

3. Consider the following triangles and answer the following questions, congruent the similar marks indicate congruent angles.



a. $\Delta XYZ \cong$ _____

b. $\Delta SRT \cong$ _____

c. $\Delta ZYX \cong$ _____

6.1.3 Tests for congruency of triangles (ASA, SAS, SSS)

At the end of this section you should be able to:

- *Describe each of the tests for congruence SAS, SSS and ASA*
- *Identify the congruence of two given triangles by using the tests for congruence SAS, SSS and ASA*

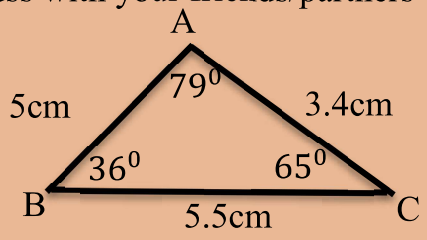
In previous discussion you have seen that, in order to say two triangles are congruent, their six corresponding part must be congruent. But, in this section, you will learn that you do not need all six pieces of information to show the triangles are congruent. By using only, the three parts of a triangle and applying the three tests for congruence (SSS, SAS, ASA), you can show whether two triangles are congruent or not.

A. Side – Side – Side (SSS)congruence test

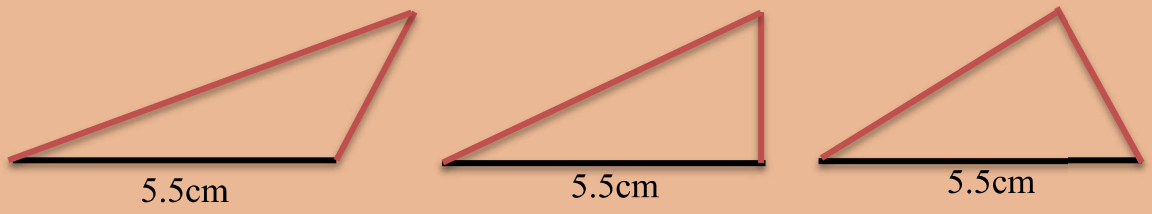
Activity 6.1.3

Read the following and discuss with your friends/partners

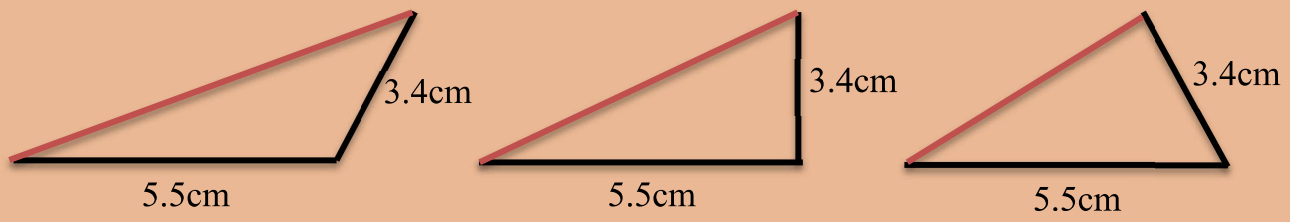
Given ΔABC,



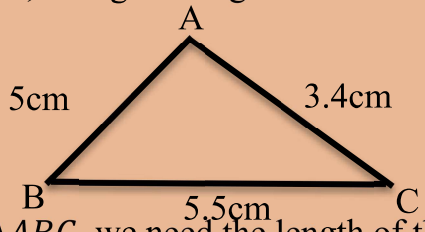
1. If only one side of ΔABC is given say 5.5cm, can you draw the exact copy of it?
   - The answer is **NO**, because you can draw a number of triangles using two another arbitrary side as follows



2. If only two sides of ΔABC is given (say 5.5cm and 3.4cm) can you draw the exact copy of it?
   - The answer is **NO**, because you can draw a number of triangles using the two sides and another arbitrary side as follows



3. If all the three sides of ΔABC is given (say BC= 5.5cm, AC= 3.4cm and AB=5cm) can you draw the exact copy of it?
   - In this case you can draw, a single triangle which the exact copy of ΔABC

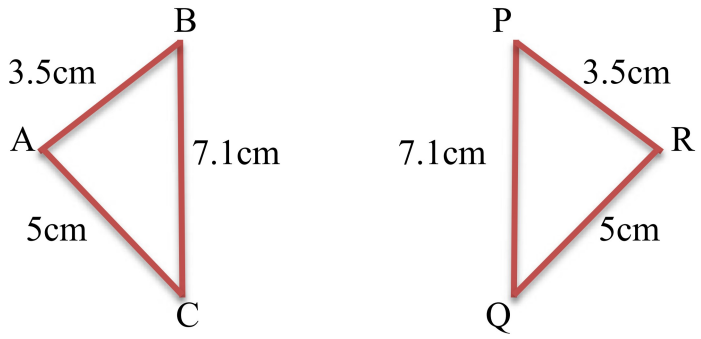


So, to draw an exact copy of ΔABC, we need the length of three sides of the triangle. We call this side – side – side congruence criterion.

Side – Side – Side (SSS) congruence test:

If the three sides of one triangle is congruent to the three corresponding sides of another triangle, then the triangles are congruent.

Example 4: examine whether the two triangles are congruent or not. If yes write the congruence relation in symbolic form

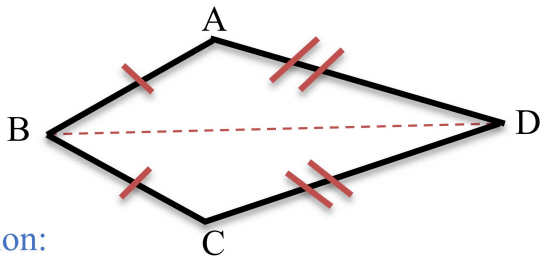


Solution:

$$AB = PR = 3.5cm \ , AC = RQ = 5cm \text{ and } BC = PQ = 7.1cm$$

- The three sides of one triangle are equal to the three sides of other triangle, so by SSS congruence test the two triangles are congruent.
- You can easily see that A corresponds to R, B corresponds to P and C corresponds to Q,
- Hence, $\Delta ABC \cong \Delta RPQ$ by SSS congruence test

Example 5: Consider the following kite



a. State the three pairs of equal parts in ΔABD and ΔCBD.
b. Is $\Delta ABD \cong \Delta CBD$? Why or why not?
c. Does $\overline{BD}$ bisect $\angle ABC$? Give reasons?

Solution:

a. Three pairs of equal parts in ΔABD and ΔCBD are

$AB = CB$…… (given)

$AD = CD$ … …(given)

$BD = BD$ … …(common side for both)

b. From a above $\Delta ABD \cong \Delta CBD$ by SSS

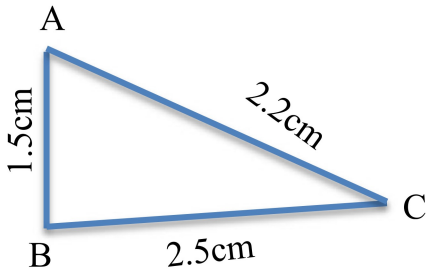
c. From b above $\angle ABD \cong \angle CBD$ …. Corresponding parts of congruent triangles

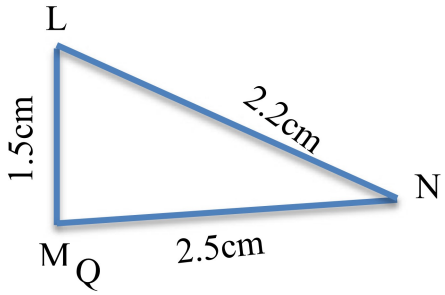
So, $\angle ABC$ is bisected by $\overline{BD}$

Exercise 6.1.3

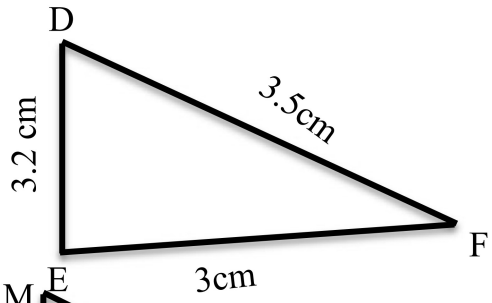
1. In the given figure below, lengths of the sides of the triangles are indicated. By applying the SSS congruence rule, state which pairs of triangles are congruent, in case of congruent triangles, write the result in symbolic form

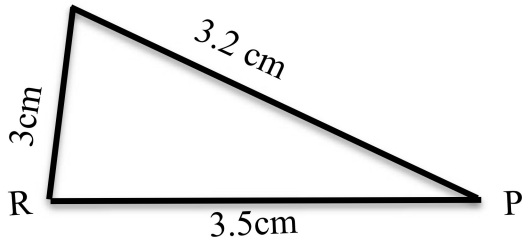
a.



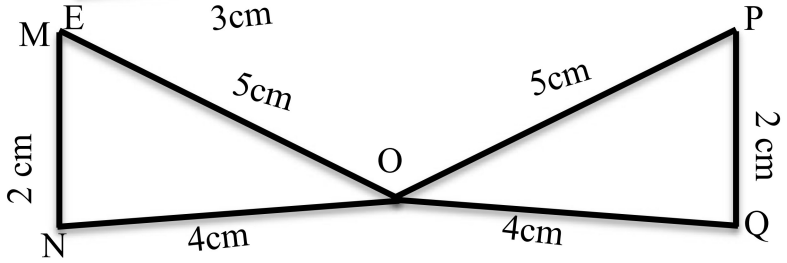


b.

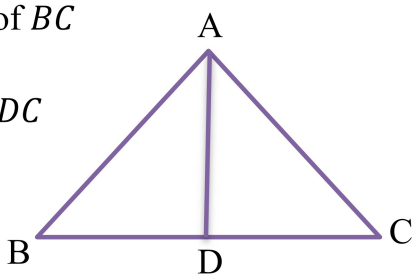




c.

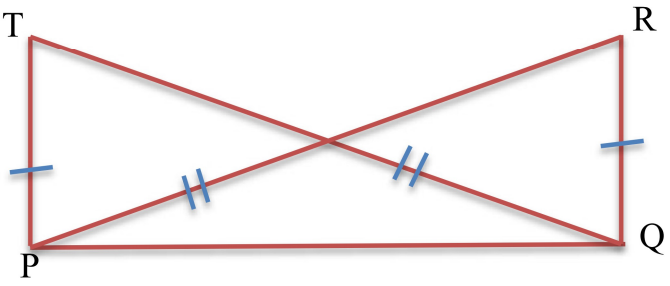


2. In the figure below $AB = AC$ and D is the midpoint of BC



   a. State the three pairs equal parts in ΔADB and ΔADC
   b. Is $\Delta ADB \cong \Delta ADC$? Give your reasons
   c. Is $< B =< C$? Why?

3. In the figure, PR=QT and PT=QR. Which of the following statements is correct?



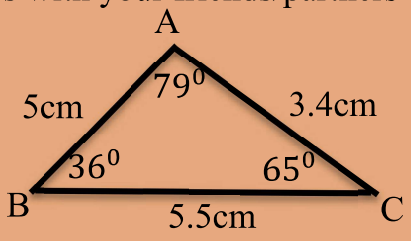
   a. $\Delta PQR \cong PQT$
   b. $\Delta PQR \cong \Delta QPT$
   c. $\Delta TPQ \cong \Delta RPQ$
   d. $\Delta PRQ \cong \Delta QTP$

B. Side – Angle – Side (SSS)congruence test

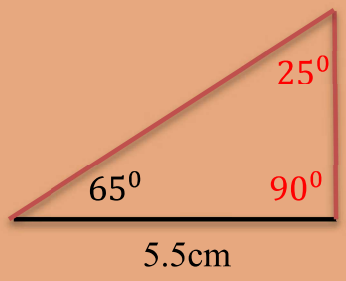
Activity 6.1.4

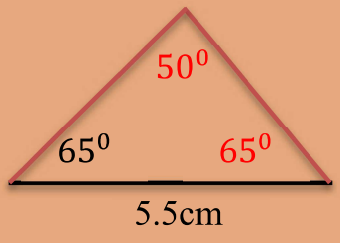
Read the following and discuss with your friends/partners

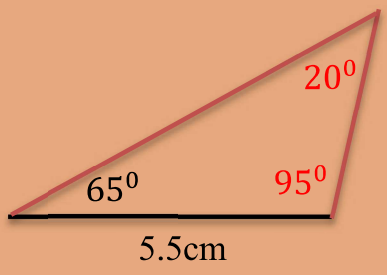
Given ΔABC,



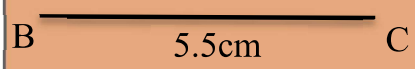
1. If only one side of ΔABC is given say 5.5cm, you can't draw the exact copy of it as you have seen in activity 5.3. can you a draw exact copy of it, if one side and one angle is given (say one side is 5.5cm and one angle is 65^0)?
   - The answer is **NO**, because you can draw a number of triangles as follows

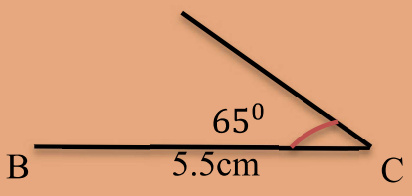


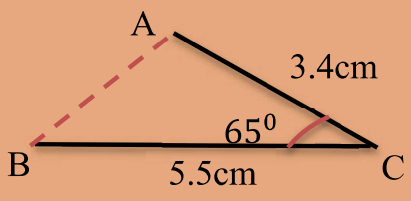




2. If two sides and the angle between these two sides is given (say BC= 5.5cm, AC= 3.4cm and the angle between BC and AC is 65^0) can you draw the exact copy of it?
   - In this case you can draw, a single triangle which the exact copy of ΔABC as follows





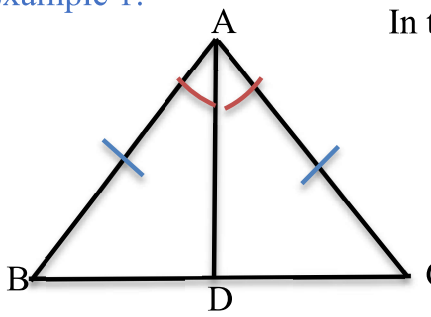


So, to draw an exact copy of ΔABC, we need the length of two sides of the triangle and the angle between the two sides. We call this side – Angle – side congruence criterion.

Side – Angle – Side (SAS) congruence test:

If the two sides and the angle included between them of a triangle are equal to two corresponding sides and the angle included between them of another triangle, then the triangles are congruent.

Example 1:



In the figure, $AB = AC$, and $\overline{AD}$ is the bisector of $\angle BAC$

i. State the three pairs of equal parts in ΔADB and ΔADC

ii. Is $\Delta ADB \cong \Delta ADC$? Give reason

iii. Is $\angle B \cong \angle C$? Give reason

iv. Is $= DC$? Give reason

Solution:

i. The three pairs of equal parts are

$AB = AC$……. (Given)

$AD = AD$……. (common sides for both)

$m(\angle BAD) = m(\angle CAD)$…. ($\overline{AD}$ bisects $\angle BAC$)

ii. yes, $\Delta ADB \cong \Delta ADC$ (by SAS)

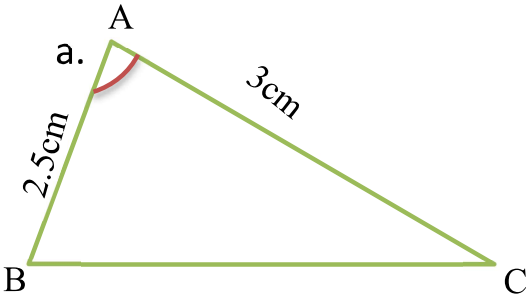
iii. yes, $\angle B \cong \angle C$, because they are corresponding parts of congruent triangles.

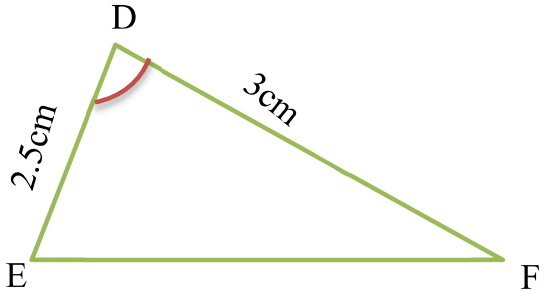
iv. yes, $BD = DC$, because they are corresponding parts of congruent triangles.

Exercise 6.1.4

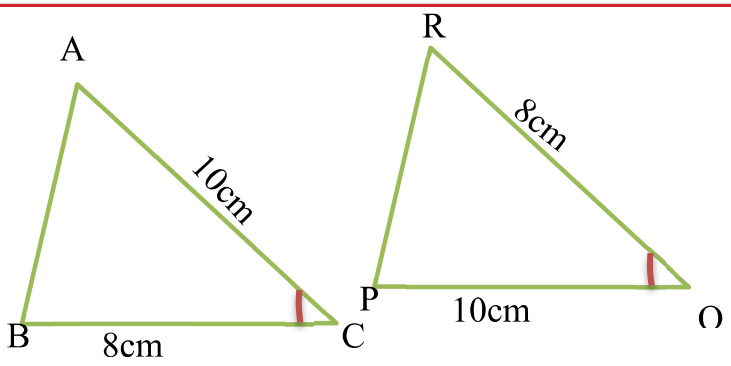
1. Which angle included between side $\overline{DE}$ and $\overline{EF}$ in ΔDEF?
2. By applying SAS congruence test, you want to establish that ΔPQR ≅ ΔFED. It is given that PQ = FE and RP = DF. What additional information is needed to be ΔPQR ≅ ΔFED?
3. In the given figure below, by applying SAS congruence test, state the pairs of congruent triangles. In case of congruent triangles, write them in symbolic form.

a.

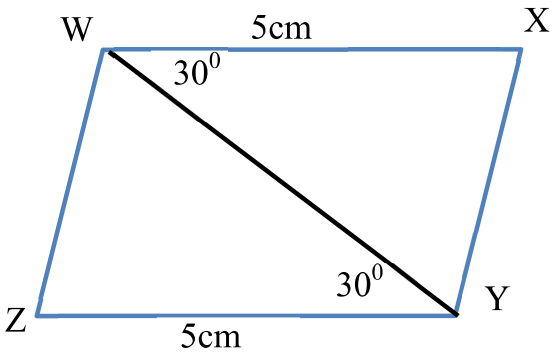




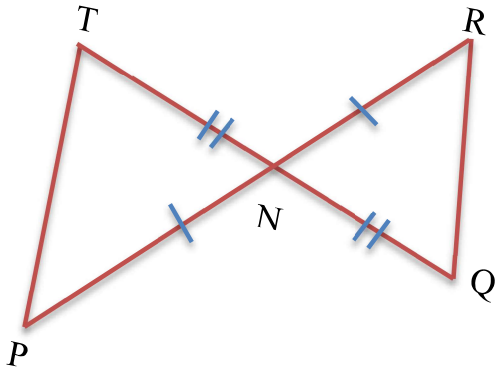
b.



c.



4. In the given figure, PN=RN and TN=QN. Which of the following statements is correct?



   a. $\Delta PNT \cong RNQ$
   b. $\Delta PNT \cong \Delta QNR$
   c. $\Delta TPN \cong \Delta RQN$
   d. $\Delta NTP \cong \Delta NQR$

C. Angle – Side – Angle (ASA)congruence test

Activity 6.1.5

Discuss with your friends/partners

1. Can you draw the exact copy of a given triangle, if you know:
   a. Only one of its angles?
   b. Only two of its angles?
   c. Two angles and any one side?
   d. Two angles and the side included between them?

Angle – Side – Angle (ASA) congruence test:

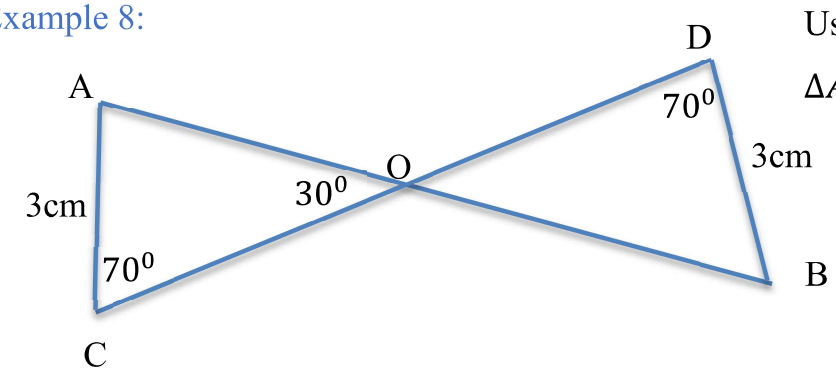
If two angles and the included side of a triangle are equal to two corresponding angles and the included side of another triangle, then the triangles are congruent.

Example 7: $\Delta ABC \cong \Delta QRP$ by ASA, and $BC = RP$ is given. what additional information is needed, to establish ASA congruence test?

Solution: For ASA congruence test, we need the two angles between the two sides BC and RP So, the additional information is

$\angle B \cong \angle R \ and \ \angle C \cong \angle P$

Example 8: Using the figure show that $\Delta AOC \cong \Delta BOD$



Solution: In $\Delta AOC, m(\angle A) + m(\angle C) + m(\angle O) = 180^0$

$m(\angle A) + 70^0 + 30^0 = 180^0$

$m(\angle A) = 80^0$

$m(\angle AOC) = m(\angle DOC) = 30^0$..... vertically opposite angles

In $\Delta DOB, m(\angle D) + m(\angle O) + m(\angle B) = 180^0$

$70^0 + 30^0 + m(\angle B) = 180^0, m(\angle B) = 80^0$

Now, $AC = DB$ and the angles that includes the sides AC and DB are also congruent

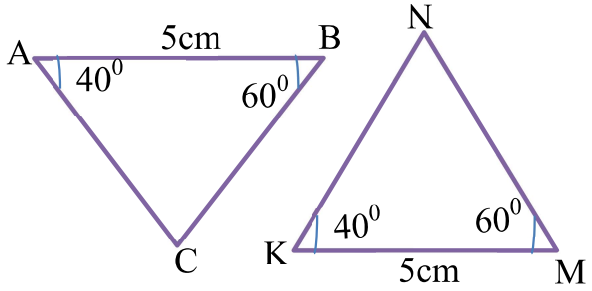
(i.e. $\angle A \cong \angle B, and \ \angle C \cong \angle D$)

So, by ASA, $\Delta AOC \cong \Delta BOD$.

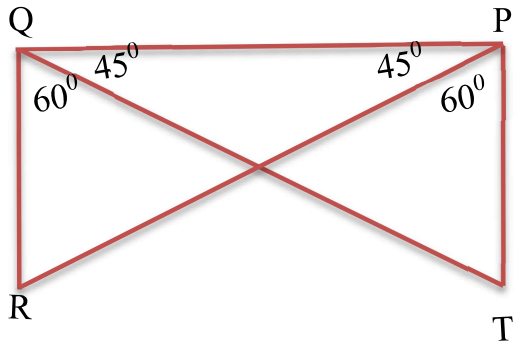
Exercise 6.1.5

1. What is the side between the angle M and N of ΔMNP.
2. You want to establish ΔDEF ≅ ΔMNP, using the ASA congruence test. You are given that ∠D = ∠M and ∠F = ∠P. What additional information is needed to establish the congruence ΔDEF ≅ ΔMNP?
3. In figure below, by applying ASA congruence test, state which pairs of triangles are congruent and write the result in symbolic form.

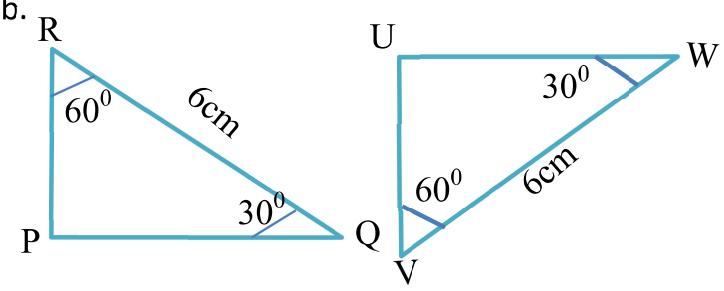
a.



c.



b.



4. Given below are measurements of some parts of ΔDEF and ΔPQR. Check whether the two triangles are congruent or not, by ASA congruence test and write it in symbolic form.

ΔDEF	ΔPQR
a. ∠D = 70⁰, ∠F = 85⁰, DF = 7 cm b. ∠D = 60⁰, ∠F = 80⁰, DF = 6 cm c. ∠E = 80⁰, ∠F = 30⁰, EF = 5 cm d. ∠D = 80⁰, ∠F = 30⁰, DF = 5 cm	• ∠Q = 70⁰, ∠R = 85⁰, QR = 7 cm • ∠P = 60⁰, ∠R = 80⁰, PR = 6cm • ∠P = 80⁰, PQ = 5 cm, ∠R = 30⁰ • ∠P = 80⁰, PR = 5 cm, ∠R = 30⁰

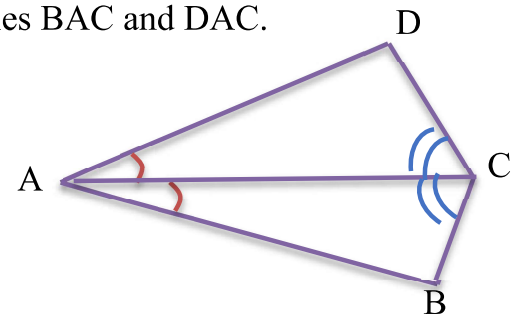
5. In the figure below, diagonal AC bisects ∠DAB and ∠DCB.

   a. State the three pairs of equal parts in triangles BAC and DAC.

   b. Is ΔBAC ≅ ΔDAC? Give reasons.

   c. Is AB = AD? Justify your answer.

   d. Is DC= CB? Give reasons



6.2. Applications

At the end of this section you should be able to:

- *apply congruency of plane figures to real life problems.*

In section you will learn some the applications of congruence of triangles.

Example 1: show that the diagonal of rhombus bisects the angles at the vertex

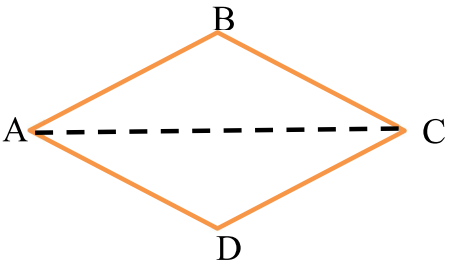
Solution: Let ABCD is rhombus, we need to show $\overline{AC}$ bisects $\angle BAD$ and $\angle BCD$

Considering ΔABC *and* ΔADC

$AB \cong AD$… All sides of rhombus are congruent

$BC \cong DC$… All sides of rhombus are congruent

$AC \cong AC$… common side for both



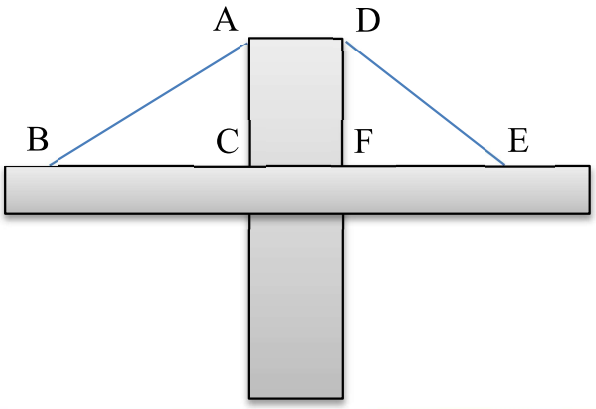
Therefore, $\Delta ABC \cong \Delta ADC$ by SSS congruence test

$\angle BAC \cong \angle DAC$… corresponding parts of congruent triangles, hence $\angle BAD$ is bisected by $\overline{AC}$.

$\angle BCA \cong \angle DCA$… corresponding parts of congruent triangles, hence $\angle BCD$ is bisected by $\overline{AC}$

In similar way you can show that $\overline{BD}$ also bisects $\angle B$ *and* $\angle D$.

Example 2: The following is a part of one side of suspension bridge on Abay river.



The length of cable AB= 5m, BC=4m, EF = 4m,

AC= 3m, what should be the length of cable DE? (assume $AC \perp BC$ and $DF \perp FE$).

Solution: consider ΔABC *and* ΔDEF

$BC = EF = 4m$ …. Given

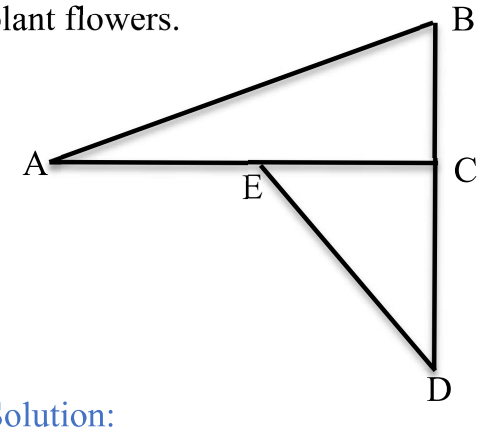
$AC = DF = 3m$….. common side

$\angle C \cong \angle F$ …. Since $AC \perp BC$ and $DF \perp FE$)

Therefore, by SAS test ΔABC *and* ΔDEF and

$DE = AB = 5m$…. Corresponding part of congruent triangles.

Example 3: Consider the following geometric figure that shows a plot of land to plant flowers.



If $= CE = 6m$, $\angle ABC \cong \angle CED$, $AE = 4m$ and $AC \perp BD$, then calculate the length of

a. CD
b. BD

Solution:

a. consider ΔABC and ΔCDE

$BC = CE$ …. Given

$\angle ABC \cong \angle CED$…. Given

$\angle ACB \cong \angle ECD$ ….. since $AC \perp BD$

The corresponding two angles and included side of the two triangles are congruent.

So, $\Delta ABC \cong \angle DEC$ by ASA.

$CD = AC$ … corresponding side of congruent triangles

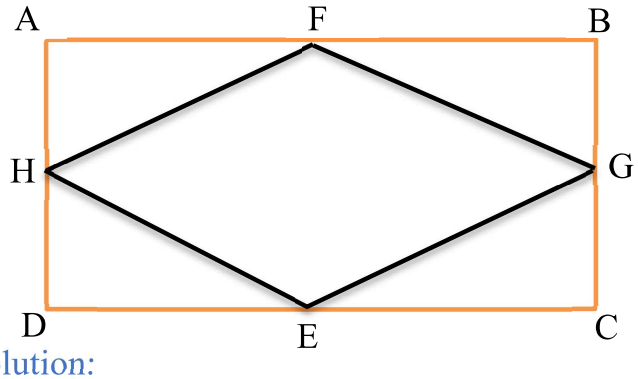
Therefore, $CD = AC = AE + EC = 4m + 6m = 10m$

b. $BD = BC + CD$

$= 6m + 10m$

$= 16m$

Example 4: In the figure ABCD is rectangle and F, G, E, and H are the mid points on side AB, BC, CD, and AD respectively.



Show that

a. $\Delta FAH \cong \Delta FBG$
b. $\Delta ECG \cong \Delta EDH$
c. $\Delta FAH \cong \Delta EDH$
d. $HF = FG = EG = EH$

Solution:

a. consider ΔFAH and ΔFBG

$FA = FB$….. since F is midpoint of AB

$AH = HD$…. Since AD = BC and, H is midpoint of AD, G is midpoint of BC

$m(\angle A) = m(\angle B) = 90^0$…. Angles of rectangle

Therefore, $\Delta FAH \cong \Delta FBG$ by SAS

b. consider ΔECG and ΔEDH

$EC = ED$….. since E is midpoint of CD

$CG = HD$…. Since AD = BC and, H is midpoint of AD, G is midpoint of BC

$m(\angle C) = m(\angle D) = 90^0$…. Angles of rectangle

Therefore, $\Delta ECG \cong \Delta EDH$ by SAS

c. consider ΔFAH and ΔEDH

$FA = ED$….. since AB =DC, and F is midpoint of AB, E is midpoint DC

$AH = HD$…. Since H is midpoint of AD

$m(\angle A) = m(\angle D) = 90^0$…. Angles of rectangle

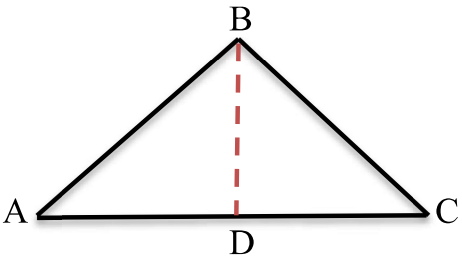
Therefore, $\Delta FAH \cong \Delta EDH$ by SAS

d. From a, b, and c you can see that $\Delta FAH \cong \Delta FBG \cong \Delta ECG \cong \Delta EDH$

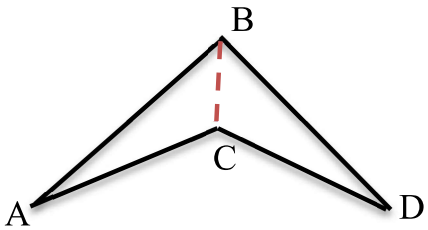
Hence, $HF = GF = GE = HE$ since the corresponding sides of congruent triangles

Exercise 6.2.1

1. Show that the diagonal of rectangle divides the rectangle in to two congruent triangles
2. In the figure below ΔABC is isosceles triangle with $AB = BC$ and BD bisects $\angle ABC$, then show that D is the mid-point of AC.



3. A portion of roof truss has the following shape



If $AB = BD$, and $\angle ABC \cong \angle DBC$, then show that $\Delta ABC \cong \Delta DBC$

Summary for unit 6

- Congruent figures are figures that have the same size and shape. Congruent figures are exact copies of one another.
- Two triangles are congruent, if their corresponding parts (angles and sides) that match one another are congruent.
- Two triangles are congruent if they are copies of each other and when you place one triangle on another, they cover each other completely.
- If ΔABC is congruent to ΔXYZ, then symbolically written as $\Delta ABC \cong \Delta XYZ$
- If $\Delta ABC \cong \Delta XYZ$, then i. $\angle X \cong \angle A$, $\angle Y \cong \angle B$ and $\angle Z \cong \angle C$
  ii. $\overline{XY} \cong \overline{AB}$, $\overline{YZ} \cong \overline{BC}$ and $\overline{XZ} \cong \overline{AC}$
- If the three sides of one triangle is congruent to the three corresponding sides of another triangle, then the triangles are congruent by Side – Side – Side (SSS) congruence rule.
- If the two sides and the angle included between them of a triangle are equal to two corresponding sides and the angle included between them of another triangle, then the triangles are congruent by Side – Angle – Side (SAS) congruence rule.
- If two angles and the included side of a triangle are equal to two corresponding angles and the included side of another triangle, then the triangles are congruent by Angle – Side – Angle (ASA) congruence rule.

Review exercise for unit 6

I. Write True if the statement is correct and False if it is incorrect

1. Congruent objects are exact copies of one another.
2. Two-line segments are congruent if they have equal lengths.
3. Two angles are congruent if their measures are equal.
4. Two triangles are congruent if the three sides of the triangle are congruent to the three corresponding sides of the other.
5. Two triangles are congruent if the three angles of one triangle are congruent to the three corresponding angles of the other.
6. If two triangles are congruent, then they have the same perimeter.
7. If two squares have equal side length, then they are congruent.
8. If $\Delta DEF \cong \Delta RTS$ then $< E \cong < R$.
9. Every triangle is congruent to itself.
10. $< Y$ is between the side YZ and XZ of ΔXYZ.

II. Choose the correct answer from the given alternatives

11. Which congruence rule do you use in the following?

    AC = DF, AB = DE, BC = EF, So, ΔABC ≅ ΔDEF

    a. SSS b. ASA c. SAS d. AAS

12. Which congruence rule do you use in the following?

    ZX = RP, RQ = ZY, ∠PRQ = ∠XZY, so, ΔPQR ≅ ΔXYZ

    a. SSS b. ASA c. SAS d. AAS

13. Which congruence rule do you use in the following?

    ∠MLN ≅ ∠FGH, ∠NML ≅ ∠GFH, ML = FG, So, ΔLMN ≅ ΔGFH

    a. SSS b. ASA c. SAS d. AAS

14. If$\Delta ABC \cong \Delta PQR$, then which of the following is not true?

    a. BC=QR b. <C=<Q c. <ABC=<PQR d. BC=PQ

15. Which one of the following is not true about congruent figures?

    A. Any two rhombuses are congruent.

    B. Any two congruent figures have the same area.

C. Any two congruent figures have the same size and shape.

D. Any two congruent circles have the same length of radius.

III. Work out

16. Show and explain using a different congruence rule for triangle why a diagonal of a square divides the square into two congruent triangles.

17. Recall that a kite is a quadrilateral with two pairs of adjacent congruent sides. Does one of the diagonals of a kite divide the kite into two congruent triangles? Show and explain your answer using congruence rule.

18. Two triangles ABC and PQR are such that; AB = 3.5 cm, BC = 7.1 cm, AC = 5 cm, PQ = 7.1 cm, QR = 5 cm and PR = 3.5 cm. Check whether the two triangles are congruent or not and explain your answer using congruence rule.

19. Given that $\Delta ABC \cong \Delta PQR$ such that $<B = (2x + 30)°$, $\angle Q = 55°$ and find the value of x.

20. Describe the rule of congruence for the two triangles given; in Δ ABC, AB = 7 cm, BC = 5 cm, $\angle B = 50°$ and in Δ DEF, DE = 5 cm, EF = 7 cm, $\angle E = 50°$.

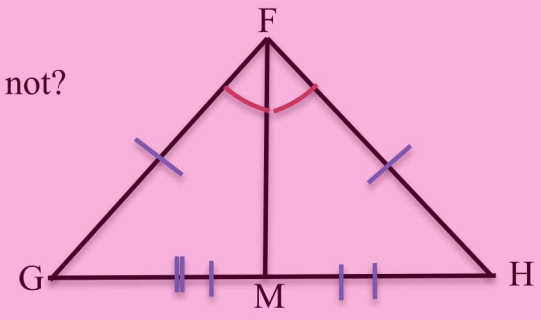
21. ΔFGH is an isosceles triangle with FG = FH and FM is one of its heights of the triangle

    a. State the three pairs of equal parts in ΔFGM and ΔFHM.

    b. Is $\Delta FGM \cong FHM$? Why or why not?

    c. Is $< G \cong < H$? Why or why not?

    d. Is $< GFM \cong \angle FHM$? Why or why not?



Unit 7
Data Handling

Learning Outcomes:

At the end of this unit, you will able to:

- ✓ *Organize data using frequency tables for a given data*
- ✓ *Construct and Interpret data from pie charts*
- ✓ *Calculate Mean, Mode, Median and range of a given data*
- ✓ *Apply the concept of data handling to organize and interpret real life problems*

Introduction

Collection of data from a group of things helps us to understand more about these things in the group. To do this the collected data should be presented systematically pictorially so as to analysis them. In this unit you will learn how to collect simple data and present them pictorially and do some calculations on them to study their nature or property.

7.1 Organization of data using frequency table

At the end of this topic, you will able to:

- ➢ *collect simple data from their environment using tally mark.*
- ➢ *organizes data in a frequency distribution table.*

Activity 7 .1.1

1. Define the word 'Data'.
2. How do you collect data?
3. The following data shows the age of students in a certain class.

15	13	13	13	15	16	14	13	14	15	12
17	15	13	13	14	15	13	16	14	13	14
15	13	16	15	16	13	12	13	15	14	14

Copy and Complete the table given below using the above information

Age of students	**Tally**	**Number of students (frequency)**
12	||	2
13		
14		
15		
16		
17		
Total number of students		

Definition 7.1: Data is a collection of facts, such as numbers, words, measurements, observations or even just descriptions of things.

You can collect data:

- ✓ by using a questionnaire.
- ✓ by making observations and recording the results.
- ✓ by carrying out an experiment.
- ✓ from records or data base
- ✓ from the internet

Note: One method of presenting data is tally chart or frequency table.

- A tally chart is a simple way of recording and counting frequencies
  - A tally chart or frequency table is a quick and easy way of recording data.
  - Frequency is the number of times a data value occurs.

Example 1: Draw tally chart or frequency table using the following data.

In a school, 40 students were asked what size of shoes does they wear. Here are the results:

33	32	34	37	37	35	38	36	37	36
36	38	35	33	36	37	38	38	31	37
36	35	37	36	39	37	36	35	38	33
34	33	37	37	38	35	34	37	39	36

Solution:

Number of friends	Tally	Frequency (Number of students)
31	|	1
32	|	1
33	||||	4
34	|||	3
35	𝍸	5
36	𝍸 |||	8
37	𝍸 𝍸	10
38	𝍸 |	6
39	||	2

Note: The tally chart or frequency table in example 1, shows the frequencies of the different size of shoe (the total number of each size of shoes is wore).

Tally marks are grouped in five (𝍸) to make counting easier:

It is easier to count in group of five tally than putting them like |||||||||.

- ✓ | represents 1 member of the group.
- ✓ 𝍸 represents 5 members of the group.

Exercise 7.1.1

1. Display the following information more clearly by drawing a tally chart or frequency table.

The following are weights in kg of 40 students in a class

40	45	45	46	44	43	45	47	46	49
44	51	47	45	44	46	46	43	44	50
48	43	45	46	44	44	47	43	44	45
45	43	45	46	44	47	45	46	44	47

2. The table below shows the favorite color of grade 7^{th} students.

White	Red	White	Yellow	Green	black	Green	Blue
Green	White	Black	Red	Yellow	Blue	Blue	Red
Yellow	Blue	White	Blue	Green	White	White	White
Yellow	Blue	Green	Green	White	Blue	Black	Red
Red	Blue	Yellow	Red	Green	White	White	Green

Display the above information in table by tally chart.

3. The following recorded data shows monthly average temperature in degree Celsius of Addis Ababa city, then display the information in a tally chart or frequency table.

21	19	19	20	20	21	22	22	23	22
22	21	20	19	19	20	21	22	22	21
19	21	21	22	20	23	22	23	24	23

7.2. Construction and Interpretation of line graphs and pie charts

At the end of this topic, you will able to:

- *Construct line graphs and pie chart to represent organized data by using the given data or by collecting data from their environment*
- *Interpret simple pie charts*

7.2.1 Line graphs

Activity 7.2.1

Discuss with your friends

1. a. Copy and complete the following table.

x	0	1	2	3	4	5
y	1	3	5	7	9	11
(x, y)						

b. Plot the above points on a Cartesian coordinate plane and connect them by straight line.

2. The following table represents score of students. Draw a line graph for the information.

score	4	5	6	7	8	9	10
Number of students	2	6	8	12	7	5	5

3. Define line graph by your own words.

Definition 7.2: Line graph is a graph that uses lines to connect individual data points on a Cartesian coordinate plane.

Note: A line graph is most commonly used to represent two related facts. It uses data points "markers," which are connected by straight lines or smooth curves.

The following points are important to making a line graph

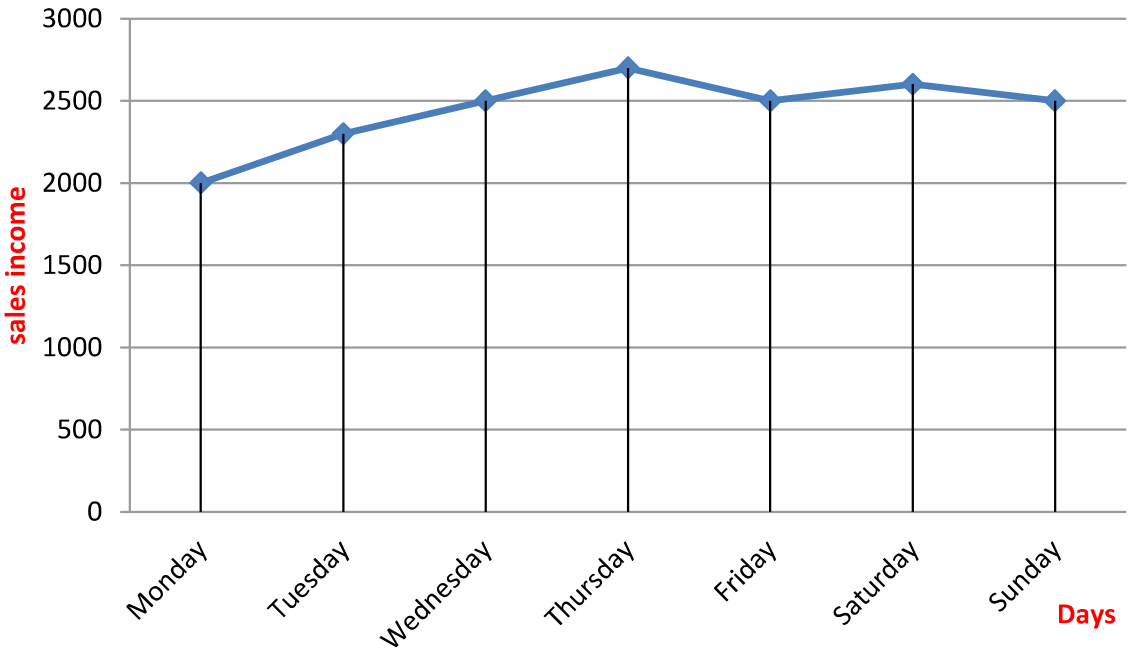
1. Construct a Cartesian coordinate plane and label an appropriate scale.
2. Make a table of data arranged in order pairs and mark the points on a Cartesian coordinate plane
3. Connect the points by a straight line or smooth curve.

Example 1: The following table shows sales income of a shopkeeper in a week. Draw a line graph of the information given below.

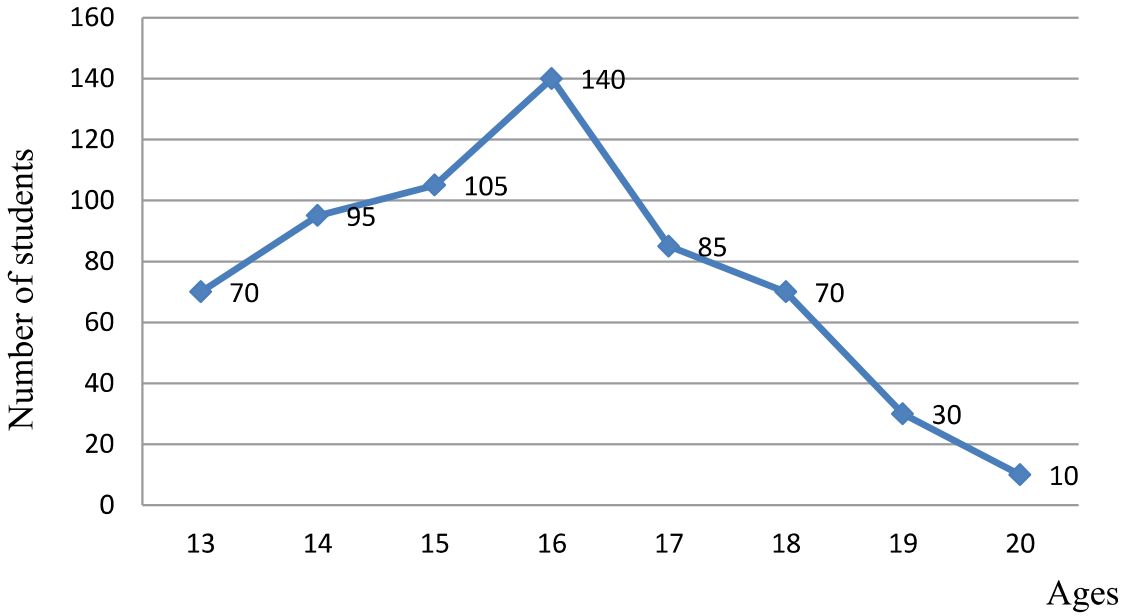
Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Selling income in Birr	2000	2300	2500	2700	2500	2600	2500

Solution: To draw the line graph of the above information

1. Draw a Cartesian coordinate plane and label the days horizontally and the selling income vertically in appropriate scale.
2. Mark the points (Monday, 2000), (Tuesday, 2300), (Wednesday, 2500), (Thursday, 2700), (Friday, 2500), (Saturday, 2600) and (Sunday, 2500)
3. Connect all points by smooth curve



Example 2: The graph below shows the number of students' in a school according to their age category.



a. What is the title of the graph?

b. Which age category has greatest number of students?

c. Which age category has least number of students?

d. Prepare a table that contains ages and the number of students.

Solution: a. Ages of students in school b. 16 c. 20

d.

Ages	13	14	15	16	17	18	19	20
Number of students	70	95	105	140	85	70	30	10

Exercise 7.2.1

1. Draw a line graph to represent each of the following data.

   a. The number of letters delivered to an office in one week

Days	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Number of Letters	10	0	5	8	12	15	10

b. The temperature in Addis Ababa at midday during the last week in April

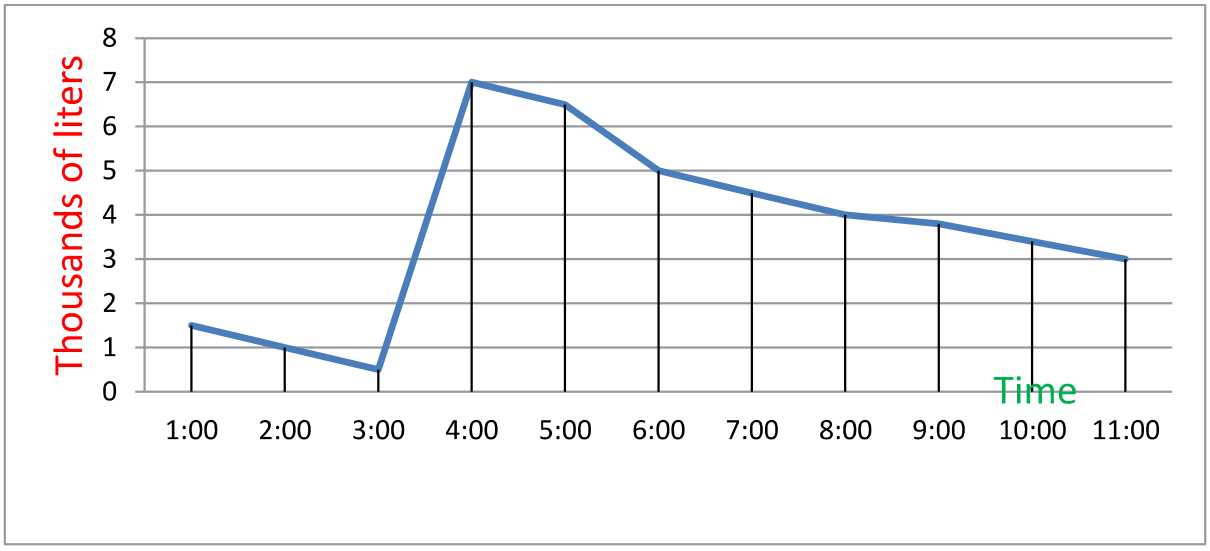
Days	Mon	Tue	Wed	Th	Friday	Sat	Sun
Temperature in ^0C	20	22	21	20	23	24	25

2. a. Copy and complete the table below gives some values between inches and centimeters. (Hint 1 inch = 2.54 centimeters).

Inches	1	2	6	10	20	30	40
Centimeters	2.54	5.08					

b. Draw line graph based on the relation between inches and centimeters.

3. The amount of petrol (in liters) in the storage tank at a garage was measured every hour between morning 1:00 and afternoon 11:30 in one day. The result is displayed in the following line graph.



a. When was the amount of petrol in the tank lowest?

b. What happened to the amount of petrol in a tank between 3:00 and 4:00?

c. What can you say about the petrol amount between 7:00 and 11:00?

7.2.2 Pie charts

Activity 7.2.2

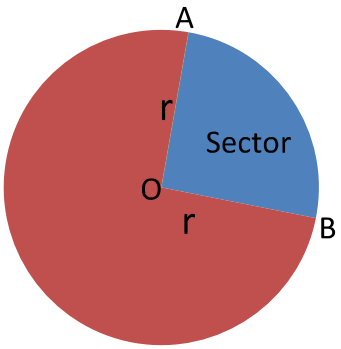
Discuss with your friends

1. Draw a circle and divide the circle in to four parts, where the central angles are 90^0. 60^0, 30^0, 180^0.
2. Look at the table given below

Age of students	Below 13	13-14	15-16	17-18	Above 18
degree of sector corresponding to number of students	20^0	244^0	54^0	32^0	10^0

Draw a pie chart to display the above data.

Circle



Note: The portion of a circular region enclosed between two radii and part of circumference is called a sector of the circle.

- r = radius
- radii is plural form of radius.
- O is the center of the circle.
- $\overset{\frown}{AB}$ is an arc of the sector AOB.

- The size of the sector is determined by the size of the angle formed by the two radii.

Note: 360^0 covers 100% of a circle.

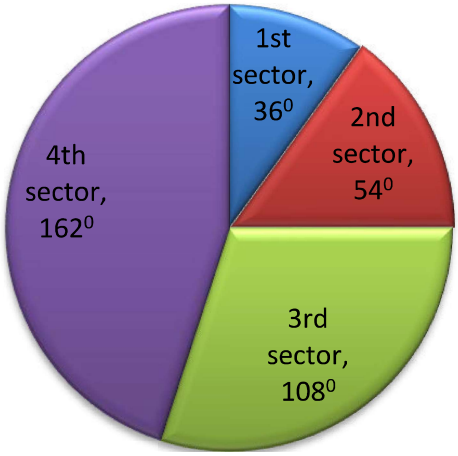
i.e 100% =360^0 ,1% =3.6^0 and $1^0 = \frac{10}{36}\%$

$$Percentage = \frac{measure\ of\ central\ angle \times total\ value}{360^0}$$

$$Measure\ of\ central\ angle = \frac{percentage \times 360^0}{total\ value}$$

Example 3: Divide the circle in to 4 sectors with $36^0, 54^0, 108^0, 162^0$

Solution: using protractor divide the circle in to sectors according to the degree measure.



Definition 7.3: Pie chart is a type of graph that represents the data in the circular graph.

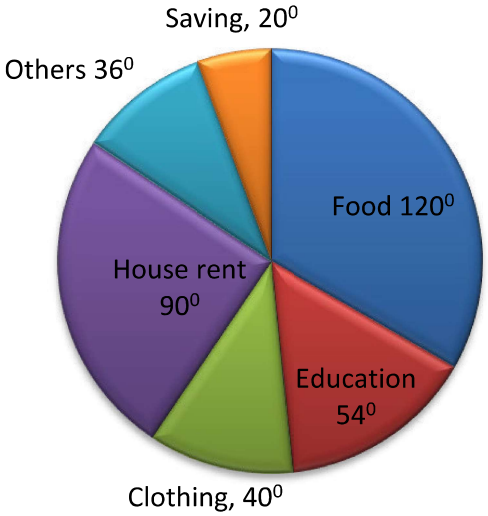
It is a very common and accurate way of representing data especially useful for showing the relations of one item with another and one item with the whole items.

Example 4: The expenditure on different budget title of a family in a month is given below in table. Draw a pie chart for the given data.

Budget title	Food	Education	Clothing	House rent	others	Saving	Total
Expenditure in Birr	2400	1080	800	1800	720	400	7200

Solution: Measure of angle $= \frac{Expenditure\ on\ the\ given\ budget \times 360^0}{Total\ expenditure}$

Budget title	Expenditure in Birr	Measure of angle of sector
Food	2400	$\frac{2400 \times 360^0}{7200} = 120^0$
Education	1080	$\frac{1080 \times 360^0}{7200} = 54^0$
Clothing	800	$\frac{800 \times 360^0}{7200} = 40^0$
House rent	1800	$\frac{1800 \times 360^0}{7200} = 90^0$
Others	720	$\frac{720 \times 360^0}{7200} = 36^0$
Saving	400	$\frac{400 \times 360^0}{7200} = 20^0$
Total	7200	360^0



Example 5: There are 1200 students in a certain school from grade 4 up to grade 8 and the number of students in percent in each grade level is given below. Draw a pie chart for the given data

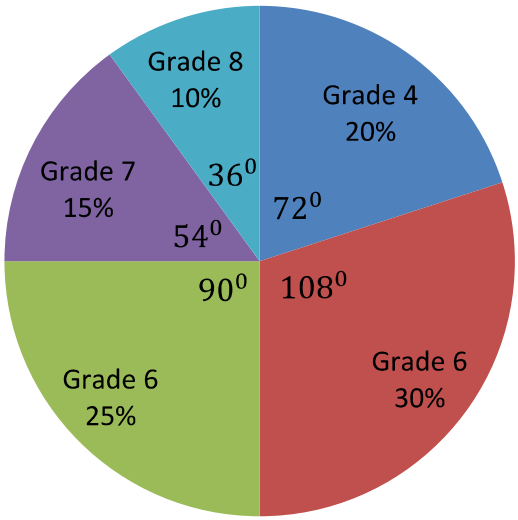
Grade	Grade 4	Grade 5	Grade 6	Grade 7	Grade 8
Number of students in %	20%	25%	15%	15%	10%

Solution:

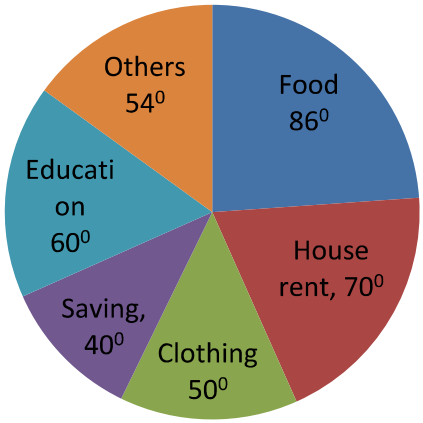
Measure of angle of sector $= \frac{percentage\ of\ students \times 360^0}{100}$

Grade	Number of students in %	Measure of angle of sectors
Grade 4	20%	$\frac{20 \times 360^0}{100} = 72^0$
Grade 5	30%	$\frac{30 \times 360^0}{100} = 108^0$
Grade 6	25%	$\frac{25 \times 360^0}{100} = 90^0$
Grade 7	15%	$\frac{15 \times 360^0}{100} = 54^0$
Grade 8	10%	$\frac{10 \times 360^0}{100} = 36^0$

Number of students from Grade 4-Grade 8



Example 6: The pie chart given below shows W/ro Tseday's expenses and saving for last month. If the monthly income was Birr 10800, then find



a. her food expenses
b. her house rent
c. her clothing expenses
d. her saving
e. her education expenses
f. for other expenses

Solution: $each\ Expenses = \frac{measure\ of\ angle\ of\ sector \times Total\ amount\ of\ budget}{360^0}$

a. Food expense $= \frac{86^0 \times 10800}{360^0} = Birr\ 2580$

b. House rent $= \frac{70^0 \times 10800}{360^0} = Birr\ 2100$

c. Clothing expense $= \frac{50^0 \times 10800}{360^0} = Birr\ 1500$

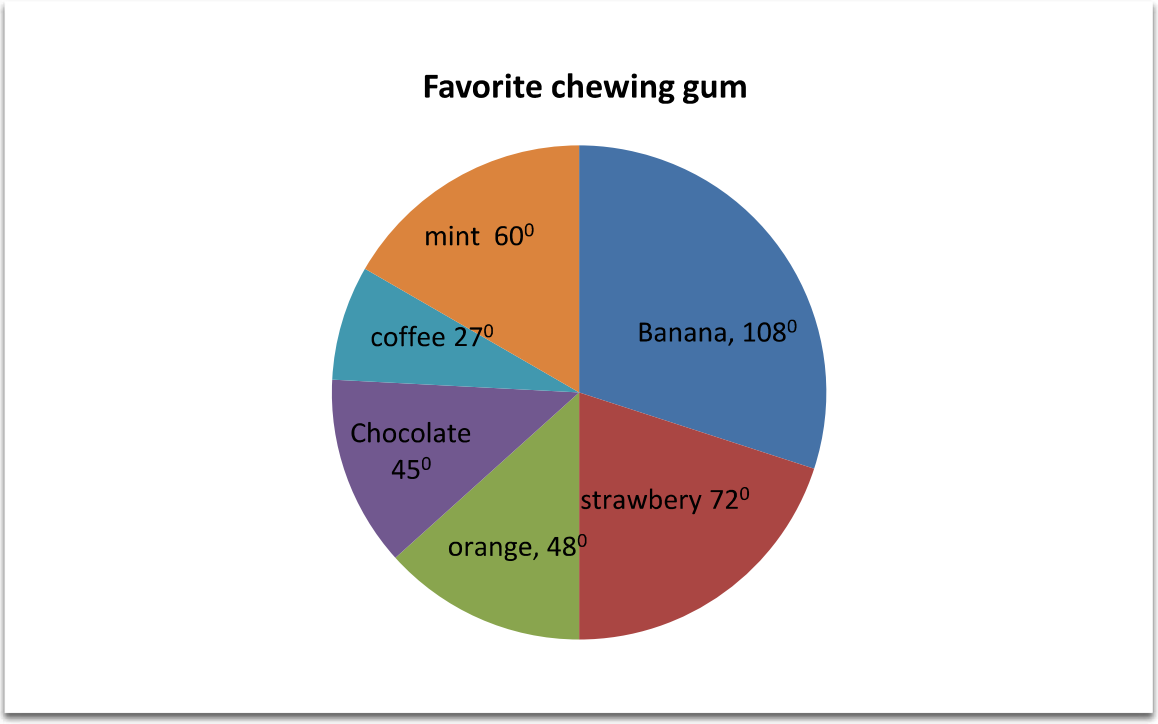
d. Saving $= \frac{40^0 \times 10800}{360^0} = Birr\ 1200$

e. Education $= \frac{60^0 \times 10800}{360^0} = Birr\ 1800$

f. For other $= \frac{54^0 \times 10800}{360^0} = Birr\ 1620$

Exercise 7.2.2

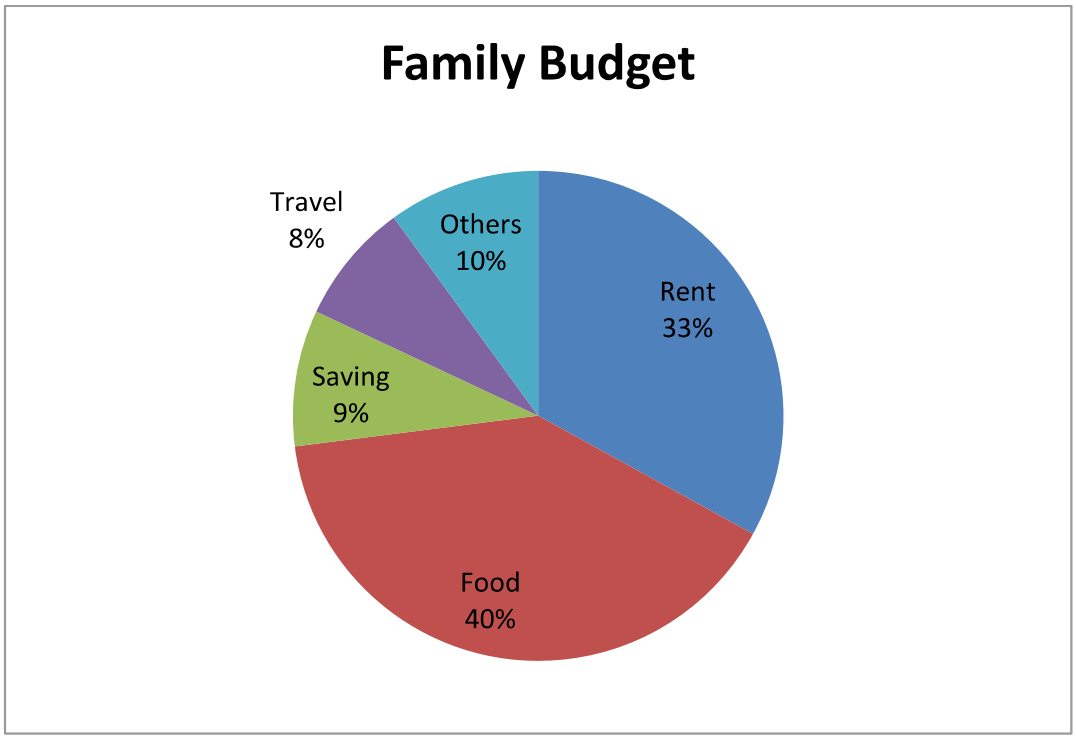
1. 1200 students were asked to name their favorite chewing gum. The results are shown in the given figure below



   a. What does the whole circle represent?
   b. What does the largest sector represent?
   c. What does the smallest sector represent?
   d. Calculate the number of students who liked

      i. Strawberry chewing gum ii. Chocolate ii. Orange

2. The following pie chart shows a family budget based on a net income of Birr 8400 per month.



a. Determine the amount spent on rent.

b. Determine the amount spent on food.

c. Determine the amount saving money.

d. How much more money is spent?

3. The budget for social development program of Woreda district is given as follows in table.

Item	Amount in Birr
Education	1,800,000
Public health	1,200,000
Community development	200,000
life skill Training for children	400,000
Total budget	3,600,000

Draw a pie chart for the given information.

7.3 The Mean, Mode, Median and Range of Data

By the end of these sections you should be able to:

- *Calculate the mean, mode, median and range of the data.*

A. Mean

Activity 7.3.1

1. Find the sum of the following numbers.

   13, 18, 13, 14, 12
2. Divide the sum of the above numbers by five.
3. How do you calculate mean?

Definition 7.4: The mean of a given data is the sum of all values divided by the number of values: mean= $\frac{sum\ of\ all\ values}{number\ of\ values}$

The mean the data $x_1, x_2, x_3, \ldots . x_n$ is

$mean(\bar{x}) = \frac{x_1+x_2+x_3+\ldots+x_n}{n}$, where $\bar{x}$ mean of data

Example1: Find the mean of the following data

a. 22, 20, 14, 12, 27

b. 100,200,120,320,150,160

Solution:

a. $mean(\bar{x}) = \frac{x_1+x_2+x_3+x_4+x_5}{5}$

$= \frac{22+20+14+12+27}{5}$

$= \frac{85}{5}$

$= 5$

b. $mean(\bar{x}) = \frac{x_1+x_2+x_3+x_4+x_5+x_6}{6}$

$= \frac{100+200+120+320+150+160}{6}$

$= \frac{1050}{6}$

$= 175$

Example 2: The mean of five numbers is 30. Four of the numbers are 32, 28, 40 and 27, then find the value of the other numbers.

Solution: Let x be the missing number

$$\text{mean } (\bar{x}) = \frac{sum\ of\ all\ values}{number\ of\ values}$$

$$30 = \frac{32+28+40+27+x}{5}$$

$$x + 127 = 150$$

$$x = 150 - 127$$

$\underline{x = 23}$, hence the fifth number is 23.

Example 3: The mean of 4 numbers is 15 and the mean of 6 other numbers is 20, then find the mean of all 10 numbers.

Solution: $mean = \frac{The\ sum\ of values}{Number\ of\ values}$

$$The\ sum\ of\ 4\ numbers = mean \times number\ of\ value = 15 \times 4 = 60$$

$$The\ sum\ of\ 6\ numbers = mean \times number\ of\ value = 20 \times 6 = 120$$

Thus, the total sum =60+120=180

Total number of data = 4+6=10

Then the mean of all ten numbers = $\frac{total\ sum}{total\ number\ of\ data} = \frac{180}{10} = 18$

Therefore, the mean of all ten numbers is 18.

Exercise 7.3.1

1. Find the mean of the following given data
   a. 10 14 16 19 25 12
   b. 28 35 70 140 160
   c. 58 40 56 55 78 80 70 45
   d. 600 750 950 800 2000
2. If the age of 8 students are 11, 12, 14, 17, 15, 13, 14 and 16, then find the mean of age of students.
3. The mean of four numbers is 70. Three of the numbers are 38, 85 and 60, then find the value of the fourth number.

4. The mean of three numbers is 18 and the mean of five other numbers is 24, then find the mean of all 8 numbers.
5. Find the value of x, so that the mean of the given data 16, 2x, 6, 10 and 4 is 8.

B. Mode

Activity 7.3.2

1. Which number occurs most frequently?

a. 6 12 10 6 23 12 6 25

b. 23 22 21 24 23 0 23 24 25 24

c. 100 500 600 700 800 900

2. Define Mode of data by your own words.

Definition 7.5: The mode of list of data is the value which occurs most frequently.

Example 4: find the mode of given data below

a. 20 30 40 20 20 30 50

b. 12 14 12 14 13 14 12

c. 25 28 50 60 70 50 60 70 90

d. 21 78 90 40 20 10 15 25 35

Solution:

a. 20 occurs more frequently than any other values of data, then mode is 20.

b. 12 and 14 occurs three times, hence three are two modes 12 andl4.

c. 50, 60 and 70 occurs two times equally, then there are three modes 50, 60 and 70.

d. Each value occurs only once, so there is no mode for the given data.

Note:

- A data that has one mode is called unimodal.
- A data that has two mode is called bimodal
- A data that has three mode is called trimodal
- If each value occurs only once, so there is no mode.

Exercise 7.3.2

1. Find the mode of the following given data below and identify it is unimodal, bimodal, trimodal and no mode.
   a. 8 11 9 14 9 15 18 6 9 10
   b. 24 15 18 20 18 22 24 26 18 26 24
   c. 35 43 39 46 51 47 38 51 43 38 40 45
   d. 123 121 119 116 130 121 131
   e. 117 119 135 121 129 139 134
   f. 7.2 6.2 7.7 8.1 6.4 7.2 5.4

C. Median

Activity 7.3.3

1. Arrange the following data in increasing order.
   a. 14 10 12 18 17
   b. 12 18 6 8 2 4 14 4 18
2. Circle the middle value of the above arrangement of numbers?

Definition 7.6: The median is the middle value when data is arranged in order of size.

To find the median of list of data:

i. Arrange data in increasing or decreasing order.
ii. Median= the middle value of arranged data.
iii. If the middle value is two data, then median is the mean of two middle data.

Example 5: Find the median of the following data

a. 4 5 6 10 14 b. 20 30 65 70 15 60 90 45

Solution:

a. The increasing order of given data is 4 5 6 10 14

Then the middle value is 6. Therefore, the median is 6.

b. The increasing order of given data is 15 20 30 45 60 65 70 90

Then the middle values are 45 and 60.

The mean of 45 and 60 is $\frac{45+60}{2} = \frac{105}{2} = 42.5$, Therefore, the median $= 42.5$

Exercise 7.3.3

1. Find the median of the following list of data

   a. 14 12 24 36 23
   b. 104 112 100 150 102 160
   c. 250 300 100 150 220 230 90
   d. 450 400 670 260 450 300 420 350 150 430

2. The salaries of 8 employees in Birr who work for a small company are listed below. What is the median salary? 8000, 9000, 5500, 10,000, 14,000, 30,000, 20,000.

D. Range

Activity 7.3.4

1. Identify the lowest and the highest value from the given data below

   a. 26 29 60 70 80 100 20 15

   b. 108 150 155 183 213

2. Find the difference between the highest and the lowest values of the above data?
3. Define range by your own words?

Definition 7.7: The range of the listed data is the difference between the highest value and the lowest value

$$Range = highest\ value - lowest\ value$$

Example 6: Find the range of the following data

a. 70 55 74 63 80 40 b. -800 -200 -600 0 -300

Solution:

a. The lowest value is 40 and the highest value is 80

$$Range = highest\ value - lowest\ value$$

$$Range = 80 - 40 = 40$$

b. The lowest value is -800 and the highest value is 0

$$Range = highest\ value - lowest\ value$$

$$Range = 0 - (-800) = 800$$

Exercise 7.3.4

1. Find the range of the following given data

a. 61 23 13 90 72 30 50

b. 46 70 85 84 83 92 94 58

c. -900 -300 -600 -460 0 -500 -250

d. 8.5 7.2 9.7 8.3 7.4 7.8 8.4

2. In a class of 20 students the highest score in mathematics exam was 94 and the lowest was 41. What was the range?

7.4 Application

By the end of these sections you should be able to:

- *apply the concept of data handling to organize and interpret real life problems.*

Example 1: The scores of 9, grade seventh students in mathematics exam is listed below. Find the mean, mode, median and range of given data.

82, 92, 78.5, 91, 92, 89, 95, 100, 86

Solution:

- $mean(\bar{x}) = \frac{x_1+x_2+x_3+x_4+x_5+x_6+x_7+x_8+x_9}{9}$

$$= \frac{82+92+78.5+91+92+89+95+100+86}{9} = 89.5$$

$$= \frac{805.5}{9}$$

- 92 occurs more frequently than any other values of data, then the mode is 92.
- The increasing order of data is

78.5 82 86 89 91 92 92 95 100

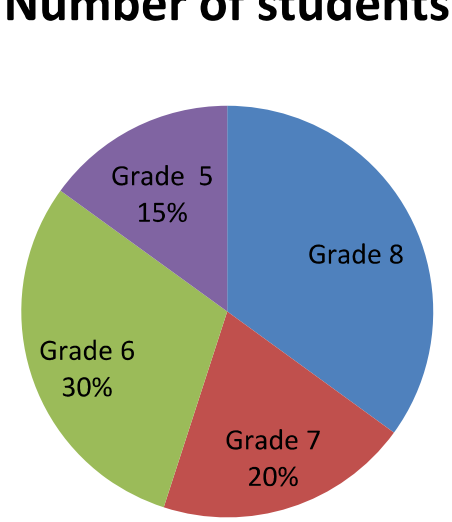
The middle value is 91, therefore the medina is 91.

- The lowest value is 78.5 and the highest value is 100

$$\text{Range= highest – lowest} = 100 - 78.5$$
$$= 21.5$$

Example 2: The pie chart shown below is the number of students in a certain school.

There are 1200 students in the school, then what is the number of students in grade 6, 7 and 8?



Solution:

$$\text{Grade 6} = 1200 \times \frac{30}{100} = 360$$

$$\text{Grade 7} = 1200 \times \frac{20}{100} = 240$$

$$\text{Grade 8} = 100\% - 30\% - 15\% - 20\% = 35\%$$
$$= 1200 \times \frac{35}{100} = 420$$

Example 3: The table shows the daily sales in Birr grocery of different categories of items for five days.

a.

Day	Mon	Tues	Wed	Thurs	Fri
Drinks	3000	4500	1500	4000	6500
Food	4000	5000	3500	3000	5000

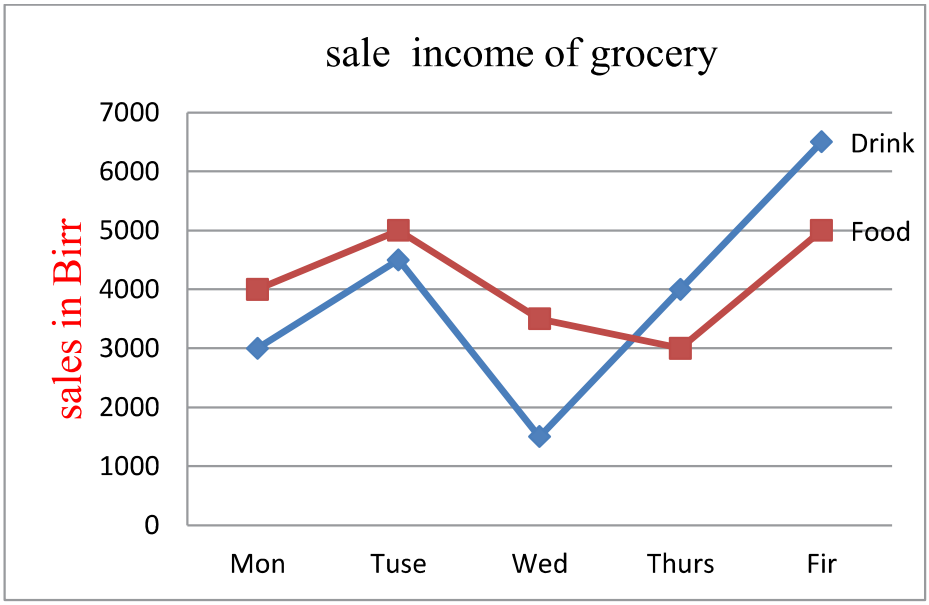
Construct a line graph for the frequency table.

b. On what days were the sales for drinks better than the sales for food?

c. What is the total earnings for food and drinks on Wednesday?

Solution:

a.



b. Thursday and Friday

c. Sale in Wednesday =Birr 1500+Birr 3500= Birr5000

Exercise 7.4.1

1. The height (in centimeters) of the members of a school football team has been listed below. 142,140,130,150,160,135,158,132. Find the mean mode, median and range of the above given data.
2. The table shows the daily earnings of a store for five days in Birr.

Day	Mon	Tues	Wed	Thurs	Fri
Earnings	300	450	200	400	650

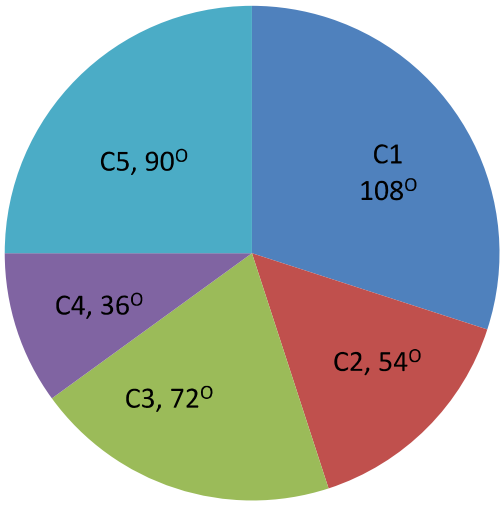
   a. Construct a line graph for the frequency table.

   b. On which days were the earnings above Birr 400.

3. The number of points scored in Ethiopian premier league of football games is listed below. Find the mean, mode, median and range of scores?

   13 18 23 9 3 18

4. 3000 students appeared for an examination from five different centers C1, C2, C3, C4 and C5 of a city. From the given pie chart, find the number of students appearing for the examination from each center.



Summary for unit 7

- A tally chart or frequency table is a quick and easy way of recording data.
- Line graph is a graph that uses lines to connect individual data points on a Cartesian coordinate plane
- The line graph is most commonly used to represent two related facts. It uses data point "markers," which are connected by straight lines or smooth curves
- Pie chart is a type of graph that represents the data in the circular Graph.
- The portion of a circular region enclosed between two radii and part of the circumference (an arc) is called a sector of the circle
- The mean of given data is the sum of all values divided by the number of values: mean= $\frac{sum\ of\ all\ values}{number\ of\ values}$
- The mode of list of data is the value which occurs most frequently.
- A data that has one mode is called unimodal.
- A data that has two mode is called bimodal
- A data that has three mode is called trimodal
- If each value occurs only once, so there is no mode
- The median is the middle value when data is arranged in order of size.
- The is $range = highest\ value - lowest\ value$

Review exercise for unit 7

I. Write True if the statement is correct and False if the statement is incorrect

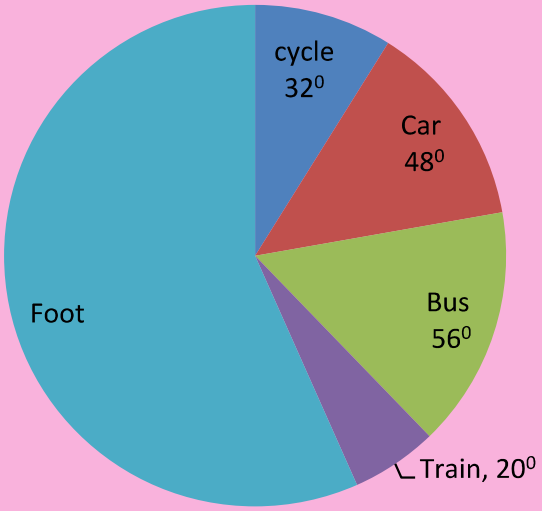
1. Mean is the difference between maximum and minimum value.
2. The mean of 1,3,5 and 7 is 5.
3. A data which has two modes is called bimodal
4. Pie chart is called circular graph.
5. 360^0 is 100% of a circle.

II. Work out

6. 1440 students were asked how they travelled to school. The pie chart shows the results of this survey;

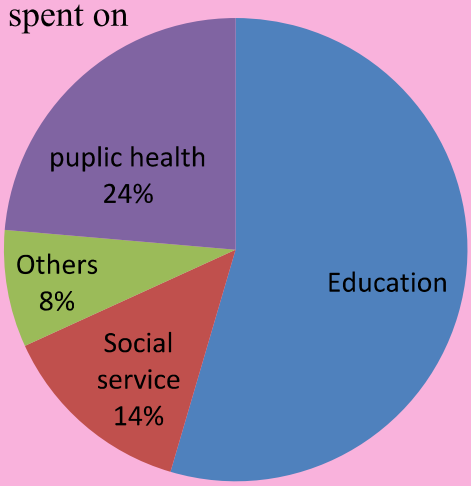
   a. how many of the students travelled to school by bus?

   b. how many students travelled on foot? In



7. The total expenditure of a sub city is Birr 720,000,000. The pie chart as shown below then how much money was spent on

   a. Education
   b. Public health
   c. Social service
   d. Others



8. If the mean of 7 numbers is 20 and the mean of 5 other numbers is 44, then find the mean of all 12 numbers?
9. Find the mean median, mode and range of each set of numbers below.
   a. 3, 4, 7, 3, 5, 2, 6, 10
   b. 8, 10, 12, 14, 7, 16, 5, 7, 9, 11
   c. 17, 18, 16, 17, 17, 14, 22, 15, 16, 17, 14, 12
10. The weights, in kilograms, of the 8 sport club members are:

    45, 53, 47, 46, 44, 46, 47, 48

    a. What is the modal weight?
    b. Calculate the mean and range of weight.
11. Students are arranged in eleven classes. The class sizes are

    23, 24, 24, 26, 27, 28, 30, 24, 29, 24, 27

    a. What is the modal class size?
    b. Calculate the mean class size.
12. Find mean, mode and median of the following numbers.
    a. 49 70 24 76 97 32 32 31
    b. 73 93 72 44 16 43 22 29
13. If the mean of x, x+3, x-5,2x and 3x is 8, then find the value x?
14. 1200 students were interviewed to find out which of five clubs they preferred. The results as follows.

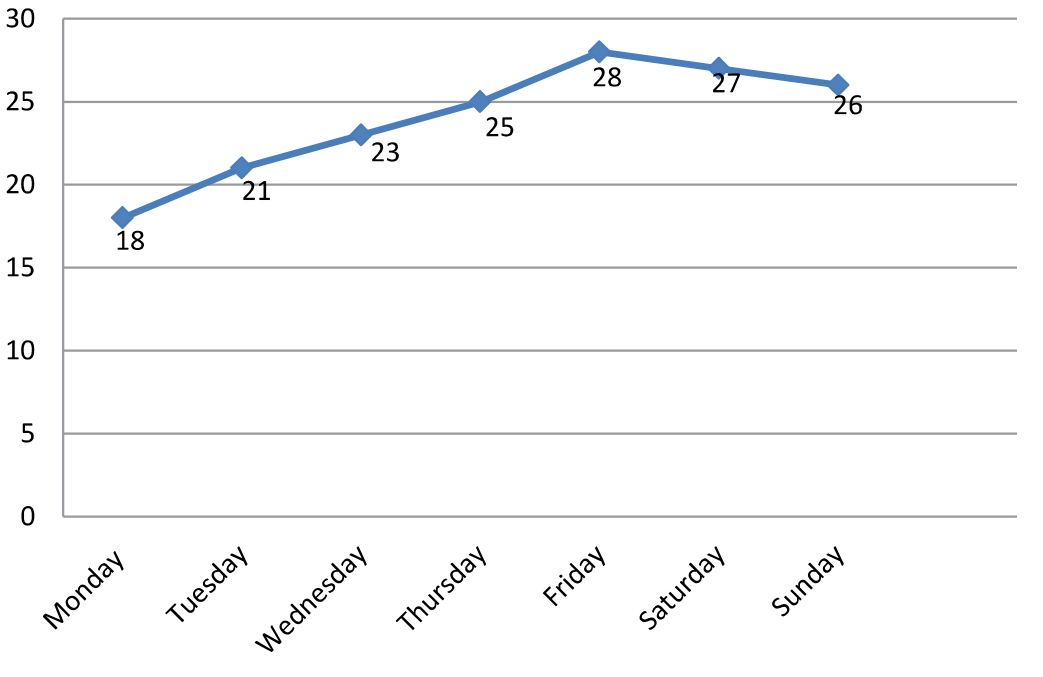
Club	Girls club	HIV club	Maths club	Sport club	Mini media club
Number of students	180	150	240	270	360

Draw a pie chart to represent the above information.

15. Use the information given to find the value of n in each of the following of numbers.

    a. 5, 7, 4, 1, n, 5: the mean is 6
    b. 3, 1, 4, 5, 4, n : the mode is 4
    c. 1, 7, 2, 1, n, 4, 3: the modes are 1 and 2
    d. 2.6, 3.5, n, 6.2: the mean is 4

16. The line graph given below shows weekly recorded Addis Ababa's temperature in degree Celsius



    a. What is the range of temperature?
    b. What is mean of temperature?
    c. When was the temperature highest?